

Step 2: Data types

Data flow

Data

Analyses

Create model

Data Quality And Insights Report



Data quality and insights report: data-quality-and-insights-report-13-6-2025_0:43:29

Target column	Type	Dataset	Date
price	Regression	madrid_rent.csv	13 de junio de 2025, 0:50 CEST



Summary

Dataset statistics

Key	Value	
Number of features	34	
Number of rows	9229	
Missing	8.23%	
Valid	91.8%	
Duplicate rows	0%	

Feature type	Count	
numeric	10	
categorical	5	
text	6	
datetime	0	
binary	12	
unknown	0	

High Priority Warnings

1 high severity warnings were detected. See the list below.

Skewed target

High



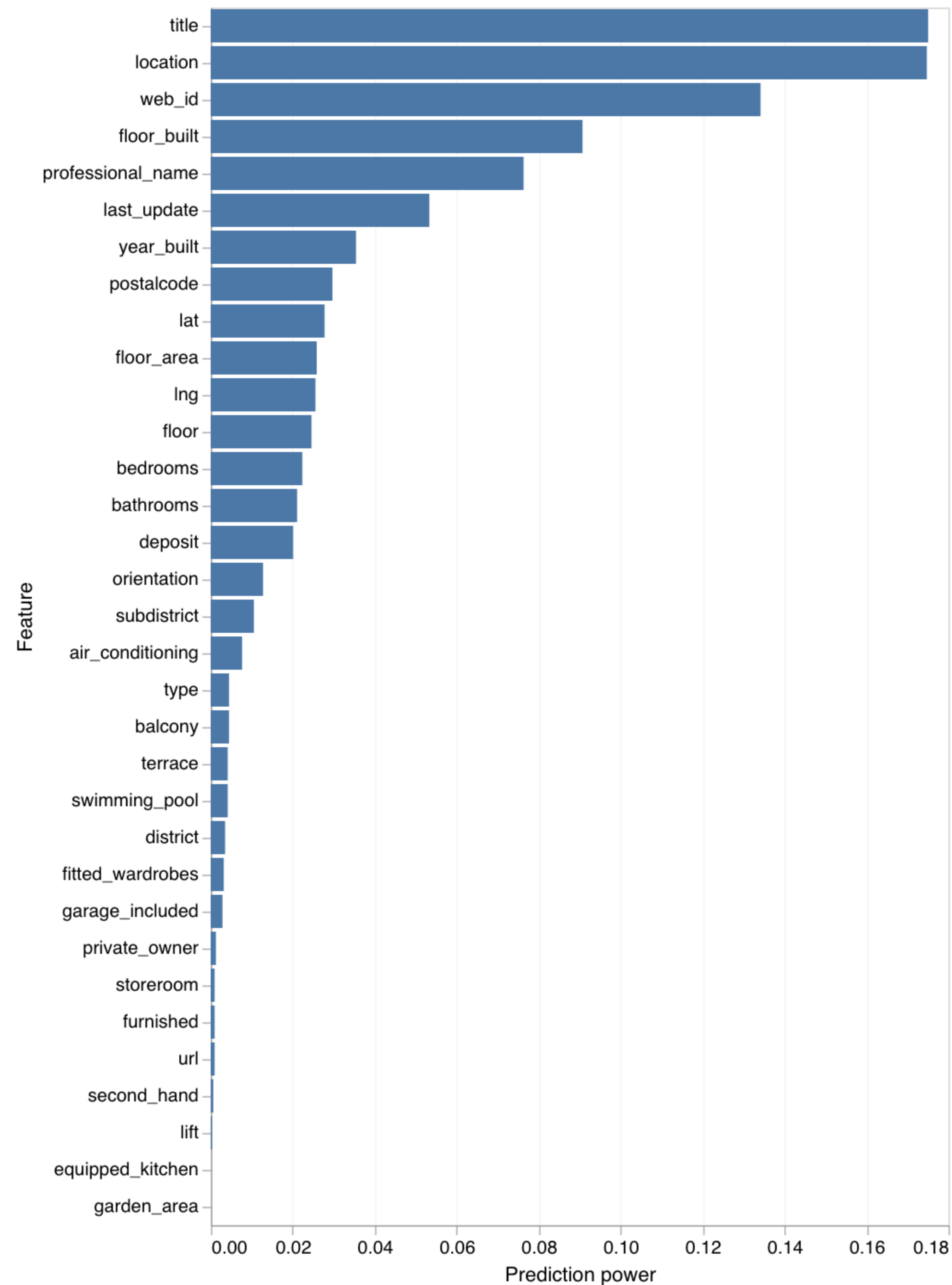
The target column is skewed and contains outliers. Because the outliers induce high errors during model training the machine learning algorithms tend to focus on them. Thus, you might get poor prediction quality for the non-outlier samples. In case you are interested in predicting extreme values well or plan to use a machine learning algorithm that has the ability to handle outlier values there is no need for further action. However, if extreme values are not the point of interest consider removing or clipping them using the Robust standard deviation numeric outliers transform under Handle outliers.

Feature summary

See a summary of the features ordered by the prediction power. Prediction power is measured by stratified splitting the data into 80%/20% training and validation folds. We fit a model for each feature separately on the training fold after applying minimal feature pre-processing and measure prediction performance on the validation data.

- The scores are normalized to the range [0,1].
- Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own.
- Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own. Although it can happen that a column is uninformative on its own but is useful in predicting the target when used in tandem with other features, a low score usually indicates the feature is redundant.
- A score of 1 implies perfect predictive abilities, which often indicates an error called target leakage. The cause is typically a column that will not be available at prediction time such as a duplicate of the target.

Feature	Prediction power	Type	Valid	Missing	High severity warnings	Medium severity warnings	
title	0.175	text	100%	0%	0	0	
location	0.175	text	100%	0%	0	0	
web_id	0.134	numeric	100%	0%	0	0	
floor_built	0.0907	numeric	100%	0%	0	0	
professional_name	0.0764	text	82.6%	17.4%	0	0	
last_update	0.0534	text	100%	0%	0	0	



Prediction power of the features. Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own. Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own.

Duplicate rows

No duplicate rows were found in the 9229 rows tested.

Note: features of type vector are ignored in duplicate rows detection

Anomalous samples

Canvas detects anomalous samples using the Isolation forest algorithm after basic preprocessing. The isolation forest associates an anomaly score to each sample (row) of the dataset.

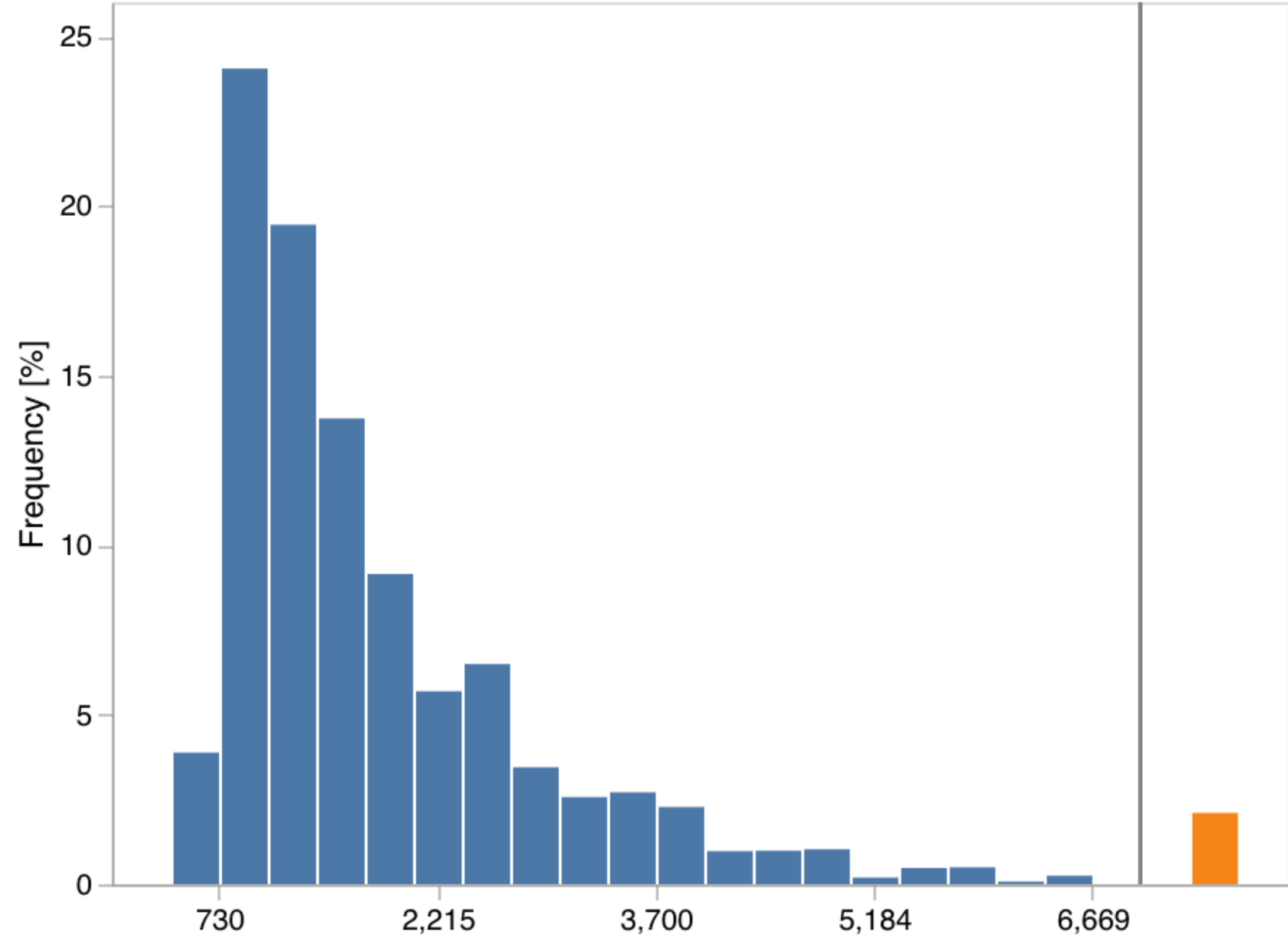
- Low anomaly scores indicate anomalous samples.
- High scores are associated with non-anomalous samples.
- Samples with negative anomaly score are usually considered anomalous and samples with positive anomaly score are considered non-anomalous.

When you look at a sample that might be anomalous, we recommend that you pay attention to unusual values. For example, you might have anomalous values that result from errors in gathering and processing the data. The following is an example of the most anomalous samples according to the Canvas's implementation of the isolation forest algorithm. We recommend using domain knowledge and business logic when you examine the anomalous samples.

Anomaly scores	web_id	url	title	type	price	deposit	private_ow
0.0979	99106978	https://www.idealista.com/en/in...	Terraced house for rent in calle ...	Terraced	2000	nan	False
0.0988	99205945	https://www.idealista.com/en/in...	Detached house for rent in calle...	Detached	15000	nan	True
0.105	99101703	https://www.idealista.com/en/in...	Terraced house for rent in Zona ...	Terraced	2000	nan	False
0.107	28696337	https://www.idealista.com/en/in...	Detached house for rent in calle...	Detached	1400	2.0	True
0.108	99218145	https://www.idealista.com/en/in...	Terraced house for rent in Palo...	Terraced	3400	nan	False
0.108	98971374	https://www.idealista.com/en/in...	Detached house for rent in Zon...	Detached	6800	nan	False

Target column

key	value	
Valid	100%	
Missing	0%	
Outliers	2.1%	
Min	400	
Max	2.5e+04	
Mean	1.94e+03	
Median	1.4e+03	
Skew	4.12	



Histogram of the target column. The red bars contain outliers and the value below them is the outliers average.

See below several samples with outlier target values.

web_id	url	title	type	price	deposit	private_owner	profession
99077965	https://www.idealista.com/en/in...	Flat / apartment for rent in plaza...	Flat	25000	1.0	False	Gilmar Ba
96643839	https://www.idealista.com/en/in...	Detached house for rent in pase...	Detached	25000	nan	False	Gilmar Pc
99292422	https://www.idealista.com/en/in...	Penthouse for rent in calle de Al...	Penthouse	20000	2.0	False	Olisson C
97728477	https://www.idealista.com/en/in...	Detached house for rent in La Fi...	Detached	19500	nan	False	Engel & V
40286753	https://www.idealista.com/en/in...	Detached house for rent in pase...	Detached	19500	nan	False	Realizza

Skewed target High

- The target column is skewed and contains outliers. Because the outliers induce high errors during model training the machine learning algorithms tend to focus on them. Thus, you might get poor prediction quality for the non-outlier samples. In case you are interested in predicting extreme values well or plan to use a machine learning algorithm that has the ability to handle outlier values there is no need for further action. However, if extreme values are not the point of interest consider removing or clipping them using the Robust standard deviation numeric outliers transform under Handle outliers.

Quick model

Quick model provides an estimate of the expected predicted quality of a model that you train on your data.

The data is split into training and validation folds where Canvas uses 80% of the samples for training and 20% of the values for validation. For classification the sample is stratified split. For a stratified split, each data partition has the same ratio of labels. For classification problems, it's important to have the same ratio of labels between the training and classification folds. Canvas trains the XGBoost model with the default hyper-parameters. It applies early stopping on the validation data and performs minimal feature pre-processing.

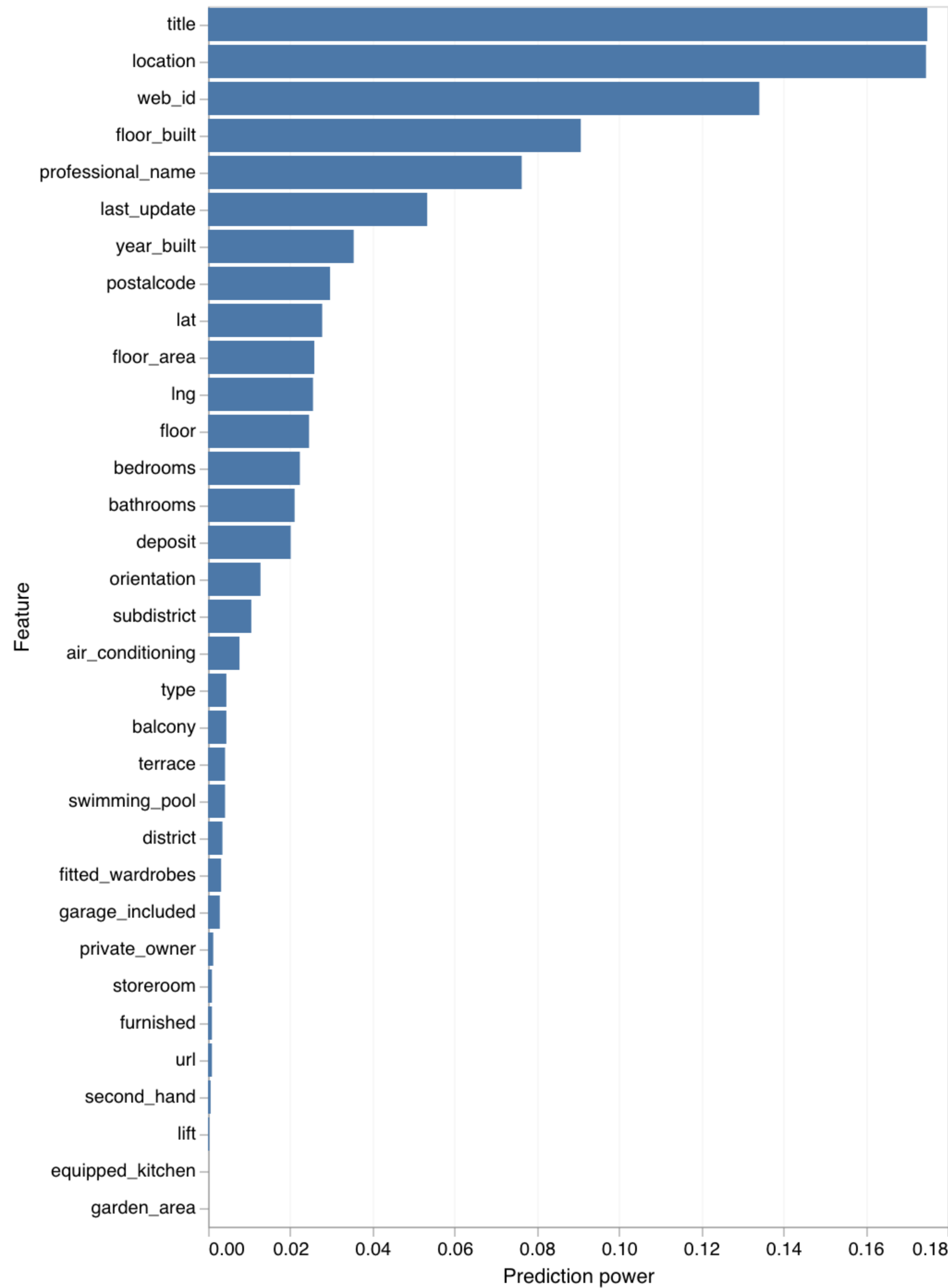
Metric	Validation scores	Train scores	
R2	0.811	0.977	
MSE	5.15e+05	5.89e+04	
RMSE	718	243	
MAE	331	168	
Max error	1.24e+04	2.49e+03	
Median absolute error	173	115	

Feature summary

See a summary of the features ordered by the prediction power. Prediction power is measured by stratified splitting the data into 80%/20% training and validation folds. We fit a model for each feature separately on the training fold after applying minimal feature pre-processing and measure prediction performance on the validation data.

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Feature	Prediction power	Type	Valid	Missing	High severity warnings	Medium severity warnings	
title	0.175	text	100%	0%	0	0	
location	0.175	text	100%	0%	0	0	
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floor_built	0.0907	numeric	100%	0%	0	0	
professional_name	0.0764	text	82.6%	17.4%	0	0	
last_update	0.0534	text	100%	0%	0	0	



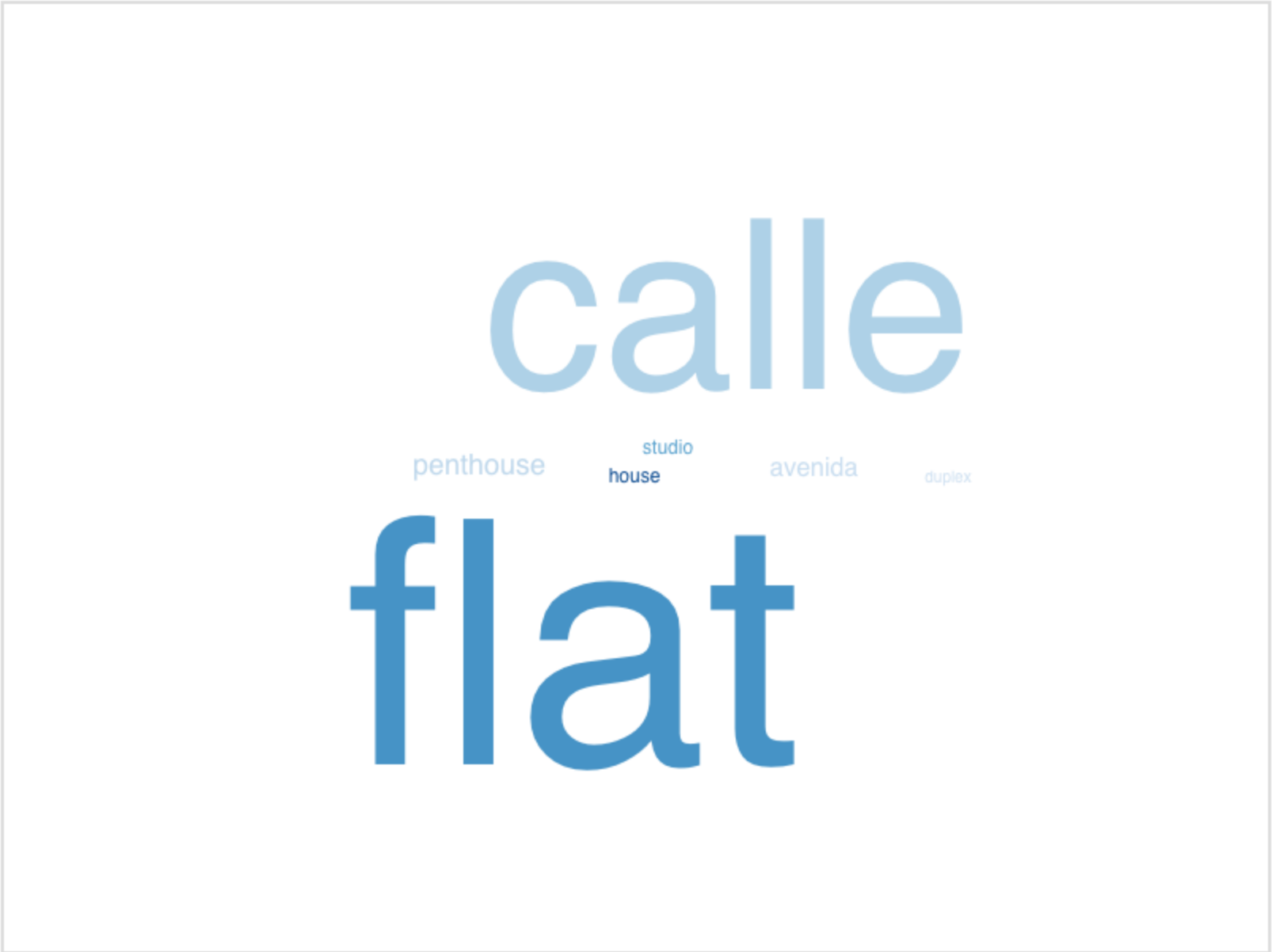
Prediction power of the features. Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own. Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own.

Feature details

title · text

key	value	
Feature name	title	
Type	text	
Prediction power	0.175	
Valid	100%	
Missing	0%	

Word	Frequency	Predict
detached	3.19%	0.034
house	5.02%	0.034
flat	84.6%	0.020
studio	4.82%	0.017
castellana	1.16%	0.017
el	1.79%	0.009
calle	58.3%	0.006



Common words. The size corresponds to word frequency and the color the predictive power. Darker words are more predictive than lighter ones.

No textual patterns found.

location · text

key	value	
Feature name	location	
Type	text	
Prediction power	0.175	
Valid	100%	
Missing	0%	

Word	Frequency	Predict
barrio	12.2%	0.071
salamanca	12.1%	0.065
la	9.38%	0.028
zona	8.79%	0.022
subdistrict	91.6%	0.014
san	8.09%	0.011
vallecas	2.78%	0.010

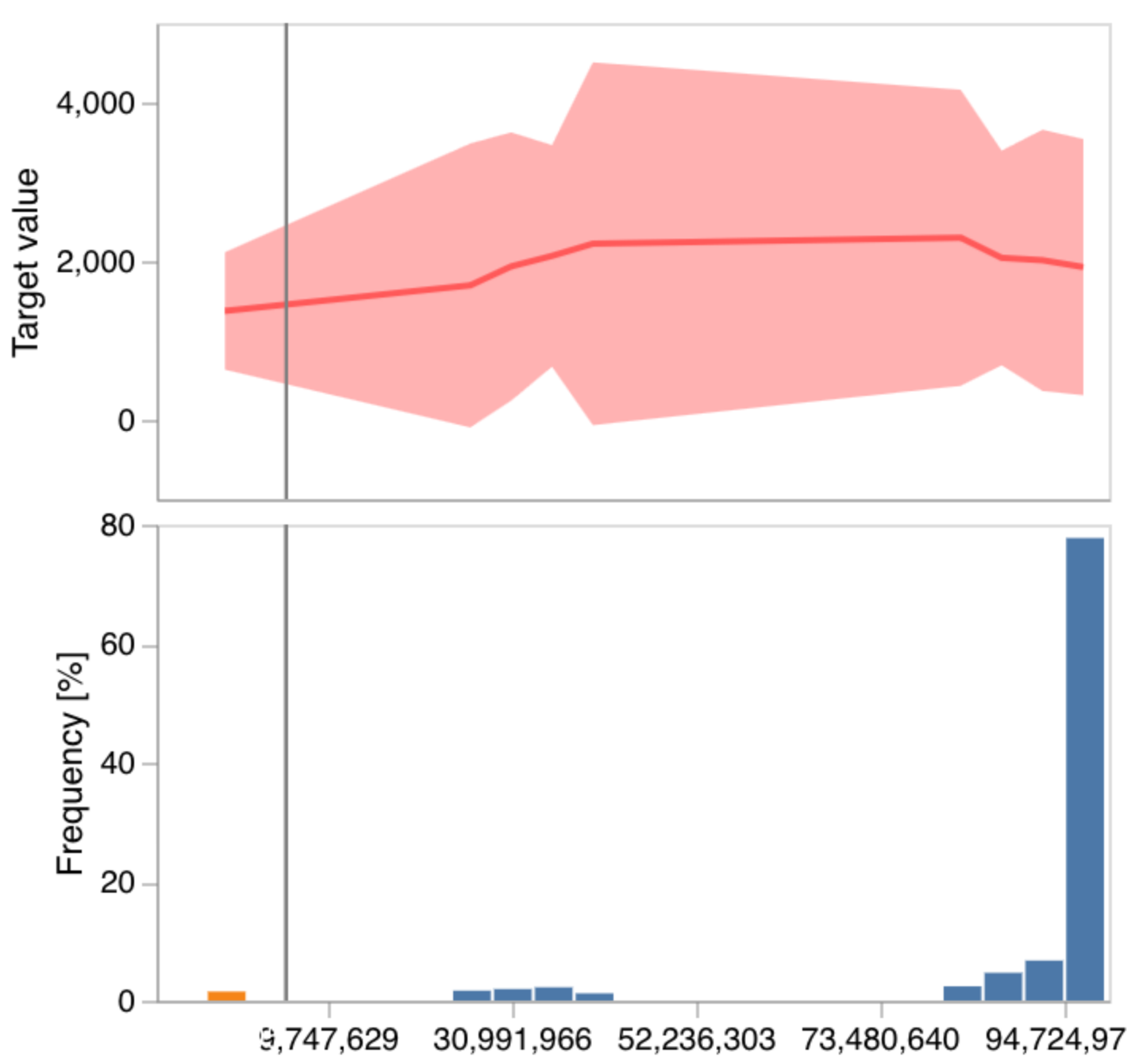


Common words. The size corresponds to word frequency and the color the predictive power. Darker words are more predictive than lighter ones.

No textual patterns found.

web_id · numeric

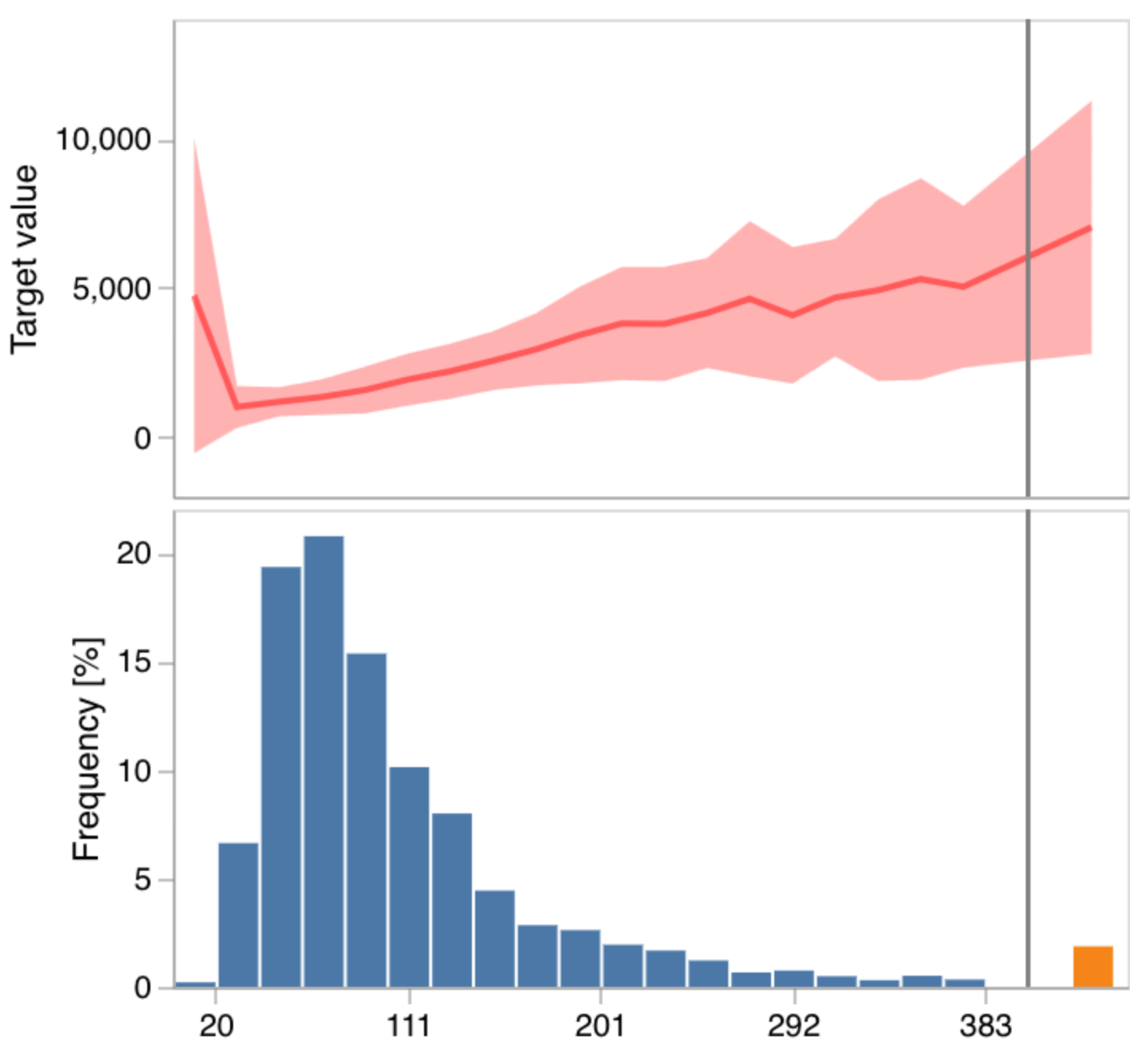
key	value	
Feature name	web_id	
Type	numeric	
Prediction power	0.134	
Valid	100%	
Missing	0%	
Outliers	1.51%	
Min	3.9e+05	
Max	9.94e+07	
Mean	9.13e+07	



Histogram of web_id with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

floor_built · numeric

key	value	
Feature name	floor_built	
Type	numeric	
Prediction power	0.0907	
Valid	100%	
Missing	0%	
Outliers	1.86%	
Min	0	
Max	990	
Mean	110	



Histogram of floor_built with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

professional_name · text

key	value	
Feature name	professional_name	
Type	text	
Prediction power	0.0764	
Valid	82.6%	
Missing	17.4%	

Word	Frequency	Predict
engel	2.3%	0.021
völkers	2.3%	0.021
home	8.92%	0.018
real	7.82%	0.017
club	5.52%	0.016
select	0.735%	0.013
estate	7.73%	0.013



Common words. The size corresponds to word frequency and the color the predictive power. Darker words are more predictive than lighter ones.

No textual patterns found.

last_update · text

key	value	
Feature name	last_update	
Type	text	
Prediction power	0.0534	
Valid	100%	
Missing	0%	

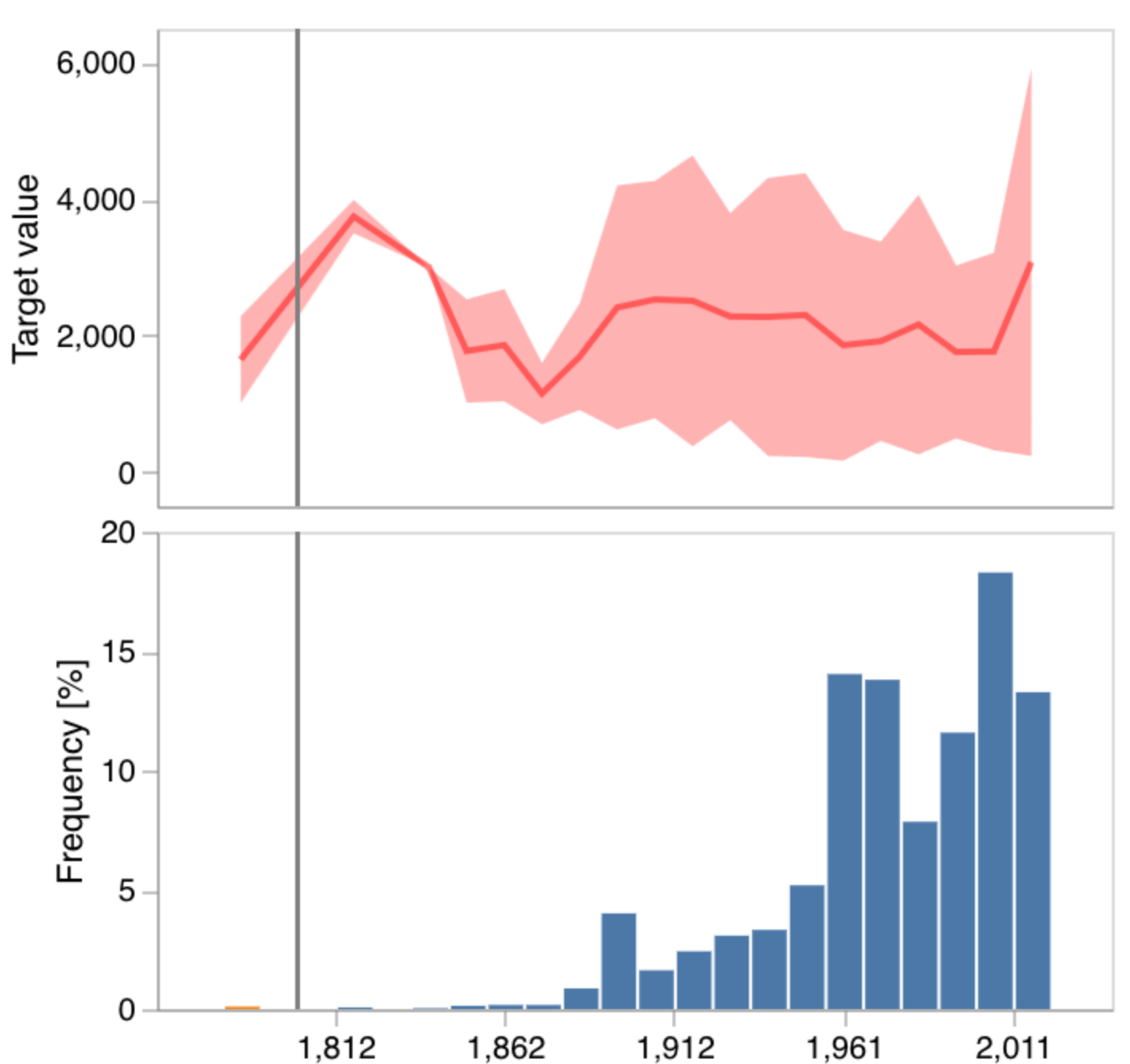
Word	Frequency	Predict
november	56.2%	0.013
february	0.596%	0.012
january	0.683%	0.011
10	0.964%	0.003
21	2.45%	0.002
december	0.347%	0.001
august	1.13%	0.001



Common words. The size corresponds to word frequency and the color the predictive power. Darker words are more predictive than lighter ones.

year_built · numeric

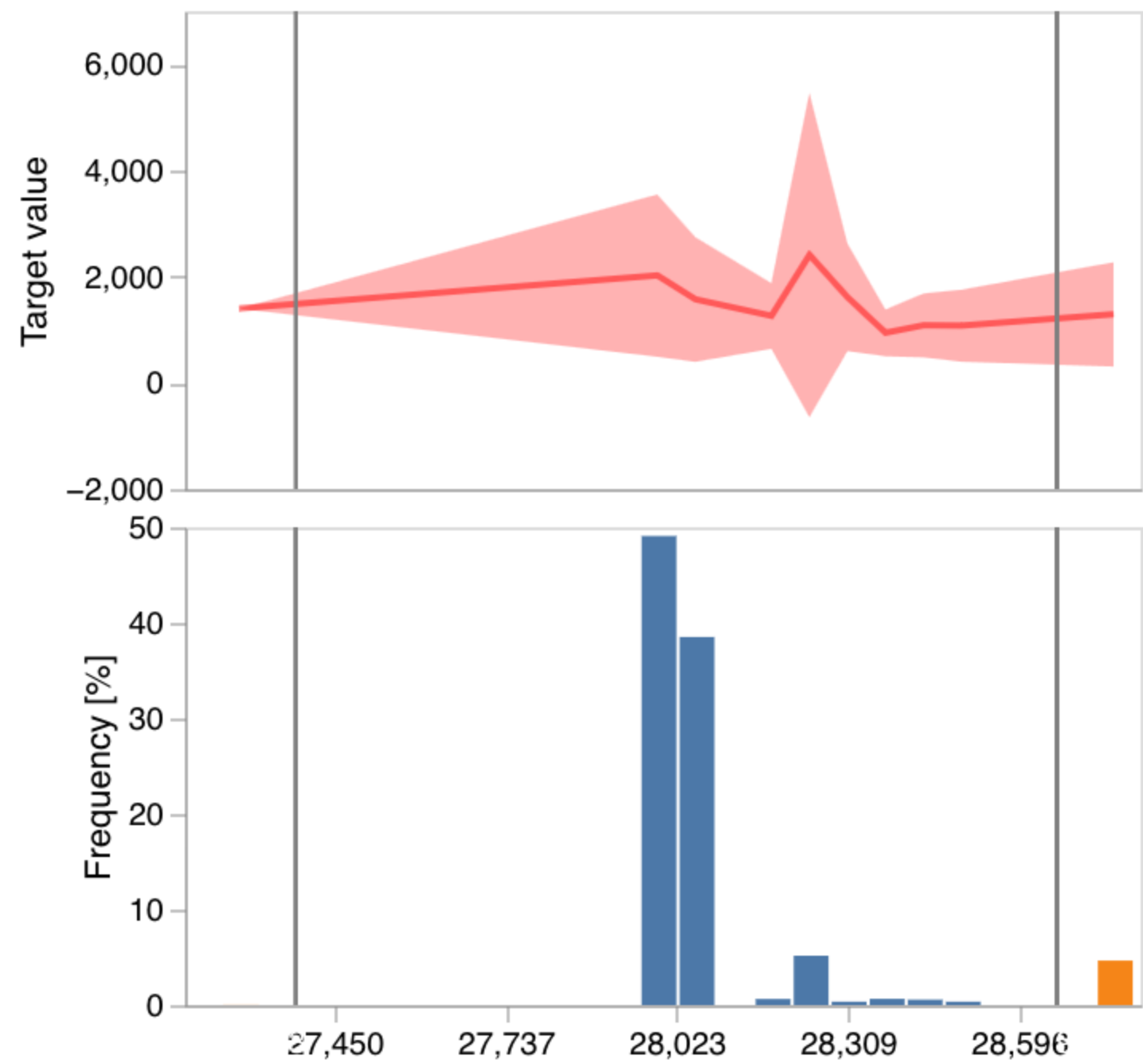
key	value	
Feature name	year_built	
Type	numeric	
Prediction power	0.0355	
Valid	31.3%	
Missing	68.7%	
Outliers	0.104%	
Min	1.8e+03	
Max	2.02e+03	
Mean	1.98e+03	



Histogram of year_built with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

postalcode · numeric

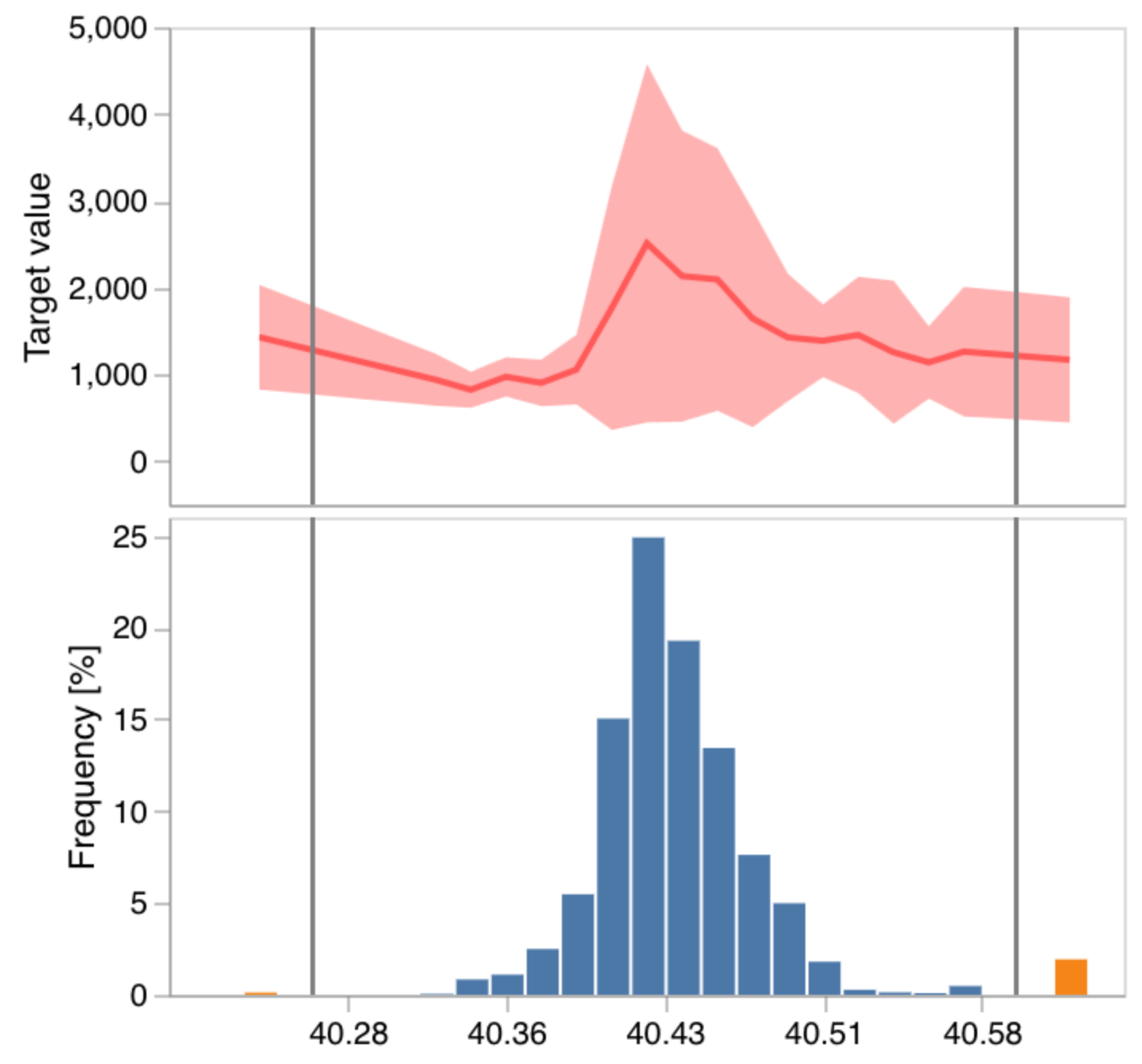
key	value	
Feature name	postalcode	
Type	numeric	
Prediction power	0.0297	
Valid	76.4%	
Missing	23.6%	
Outliers	4.67%	
Min	9.62e+03	
Max	9.5e+04	
Mean	2.81e+04	



Histogram of postalcode with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

lat · numeric

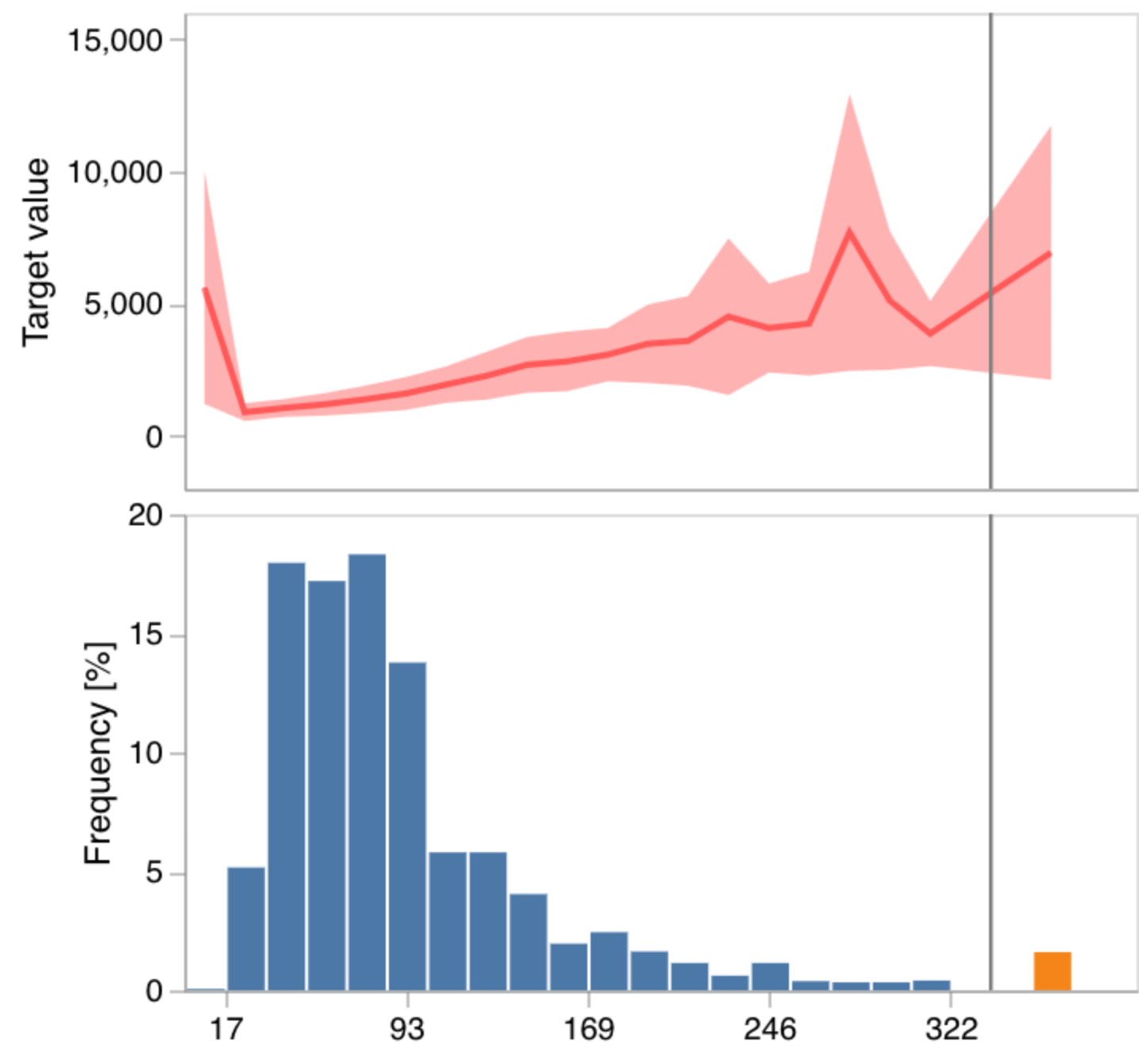
key	value	
Feature name	lat	
Type	numeric	
Prediction power	0.0278	
Valid	100%	
Missing	0%	
Outliers	2.05%	
Min	-34.5	
Max	56.3	
Mean	40.4	



Histogram of lat with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

floor_area · numeric

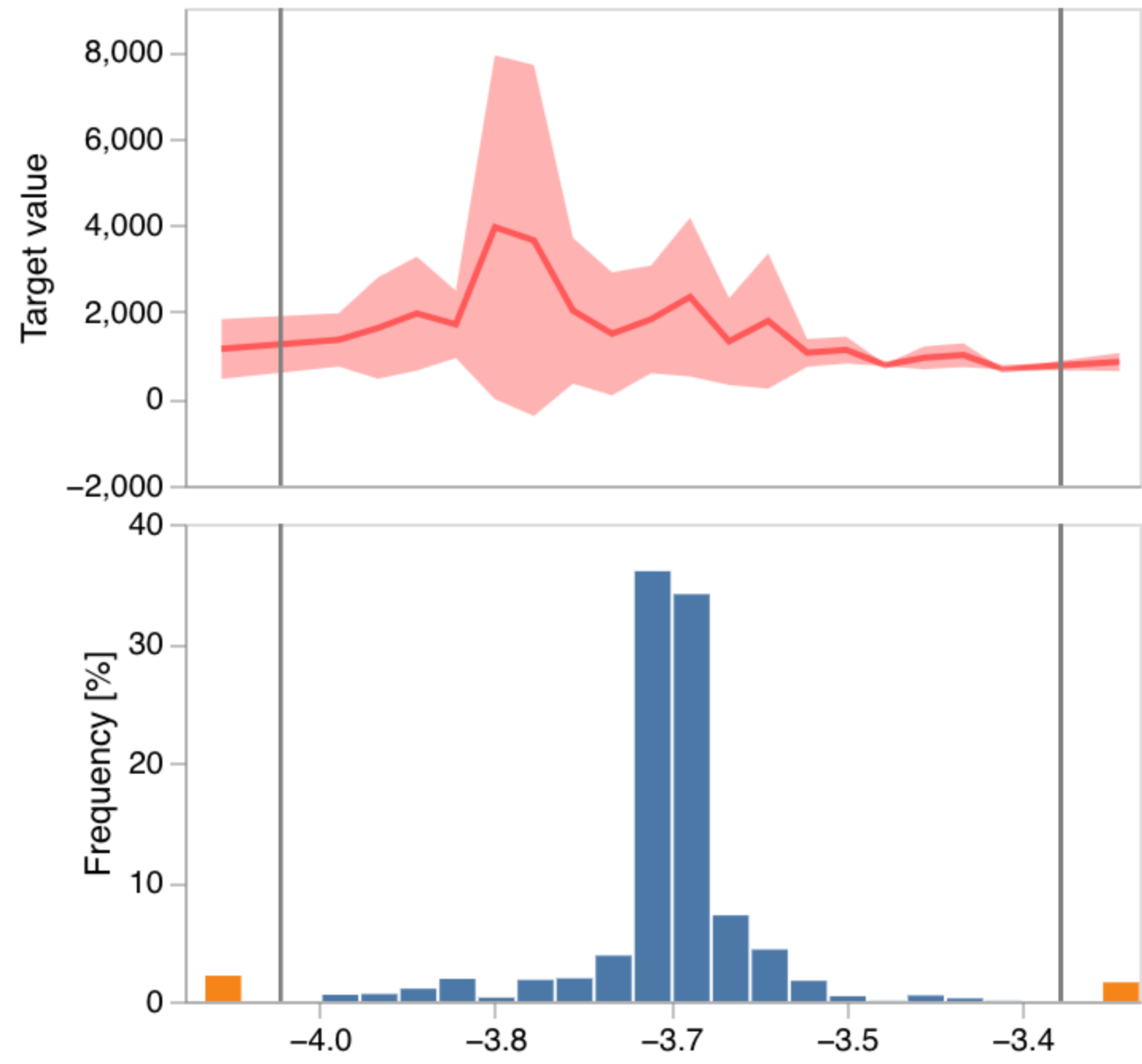
key	value	
Feature name	floor_area	
Type	numeric	
Prediction power	0.0259	
Valid	42.7%	
Missing	57.3%	
Outliers	1.6%	
Min	0	
Max	995	
Mean	94.4	



Histogram of floor_area with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

lng · numeric

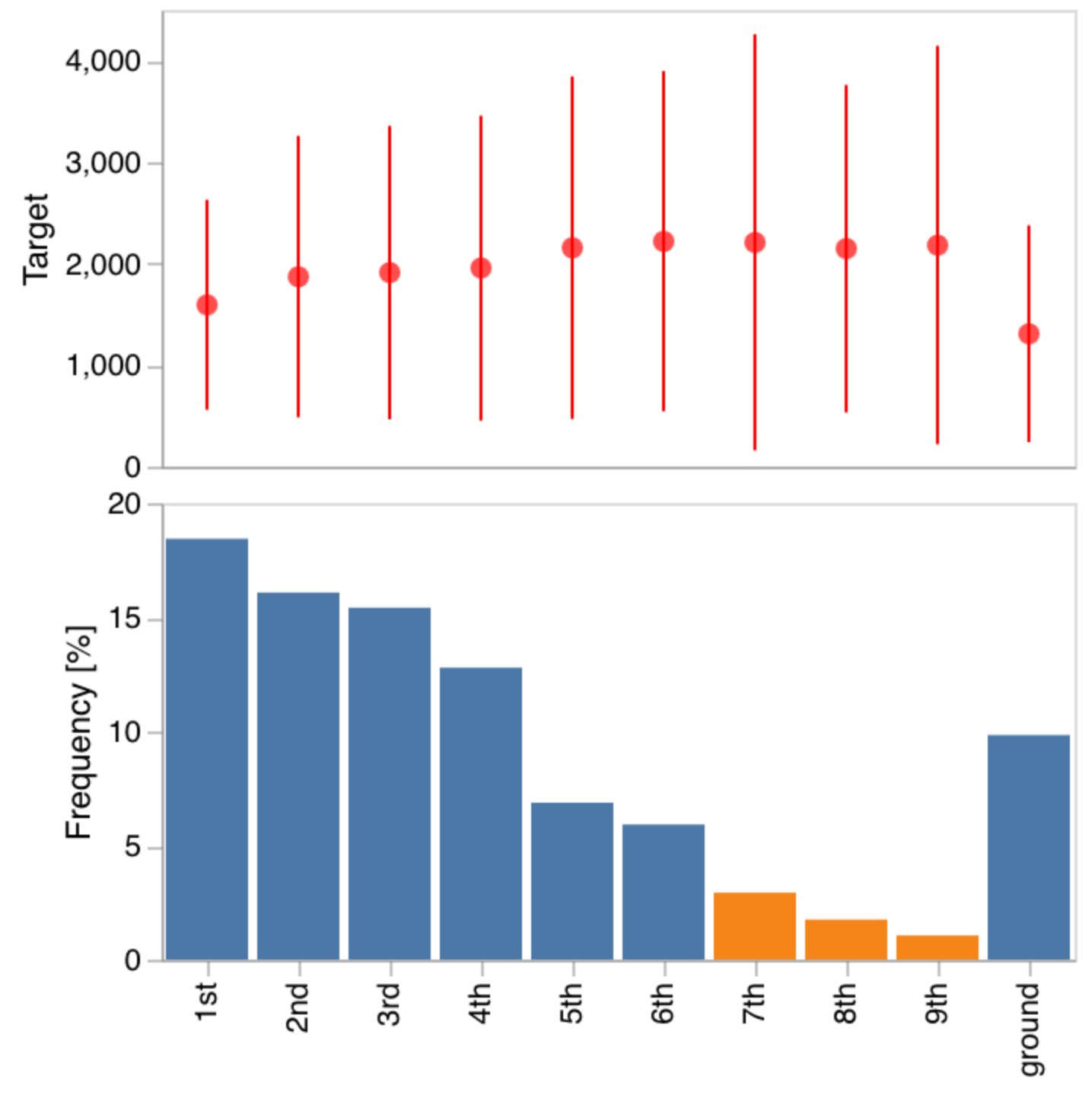
key	value	
Feature name	lng	
Type	numeric	
Prediction power	0.0256	
Valid	100%	
Missing	0%	
Outliers	3.71%	
Min	-122	
Max	21.8	
Mean	-3.73	



Histogram of lng with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

floor · categorical

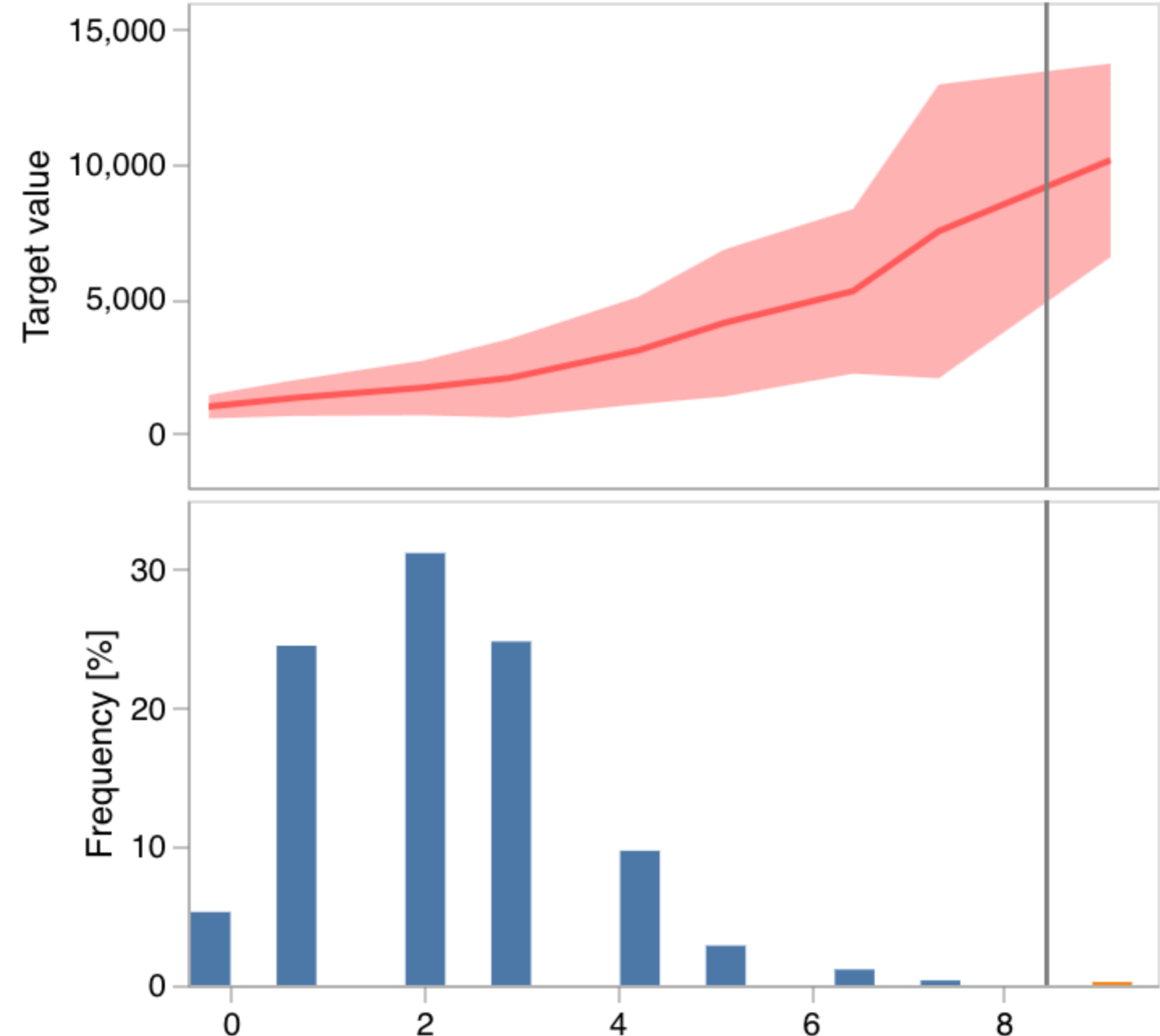
key	value	
Feature name	floor	
Type	categorical	
Prediction power	0.0246	
Valid	96.5%	
Missing	3.48%	



Histogram of the frequent values of floor with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

bedrooms · numeric

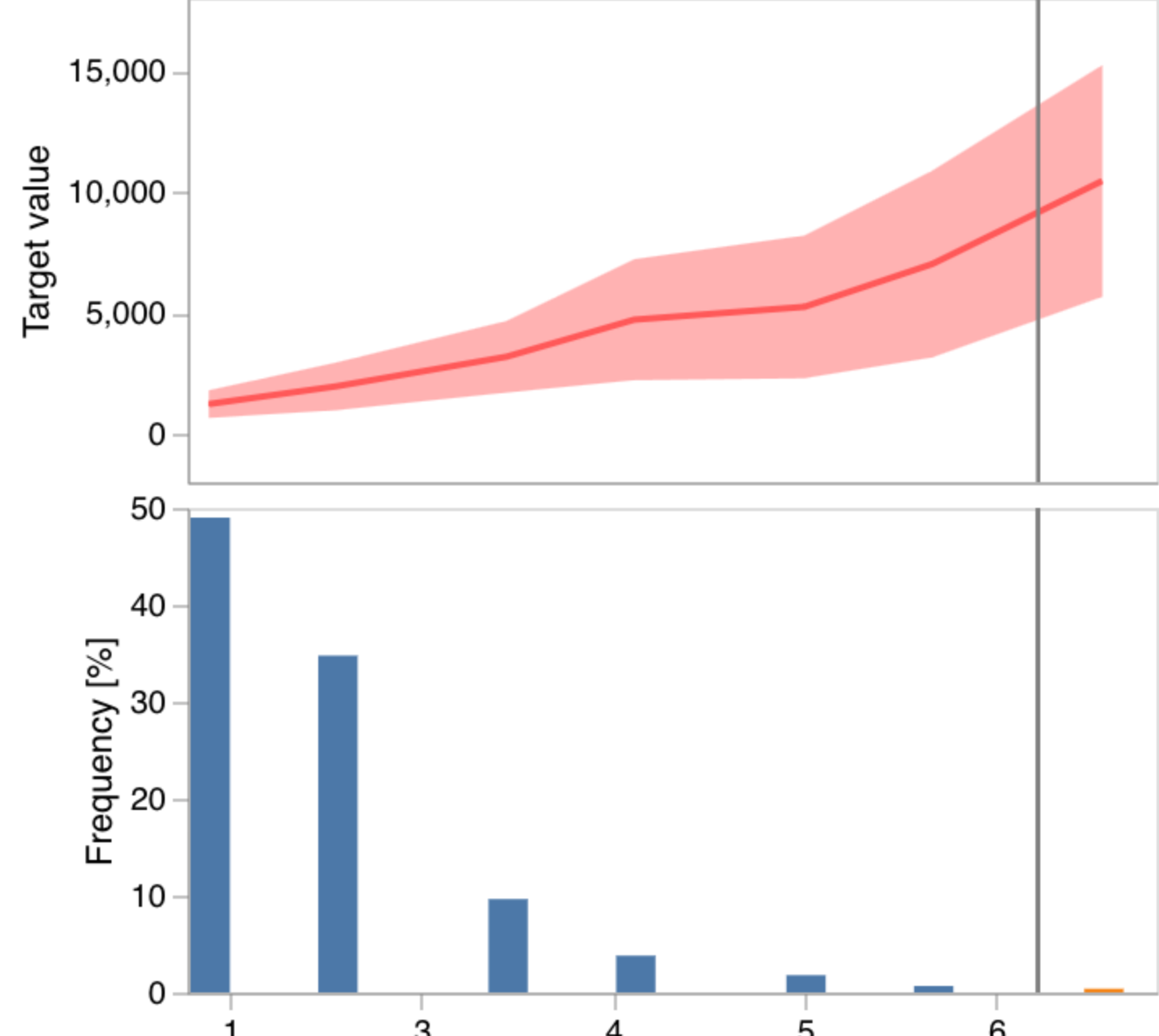
key	value	
Feature name	bedrooms	
Type	numeric	
Prediction power	0.0224	
Valid	100%	
Missing	0%	
Outliers	0.228%	
Min	0	
Max	25	
Mean	2.26	



Histogram of bedrooms with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

bathrooms · numeric

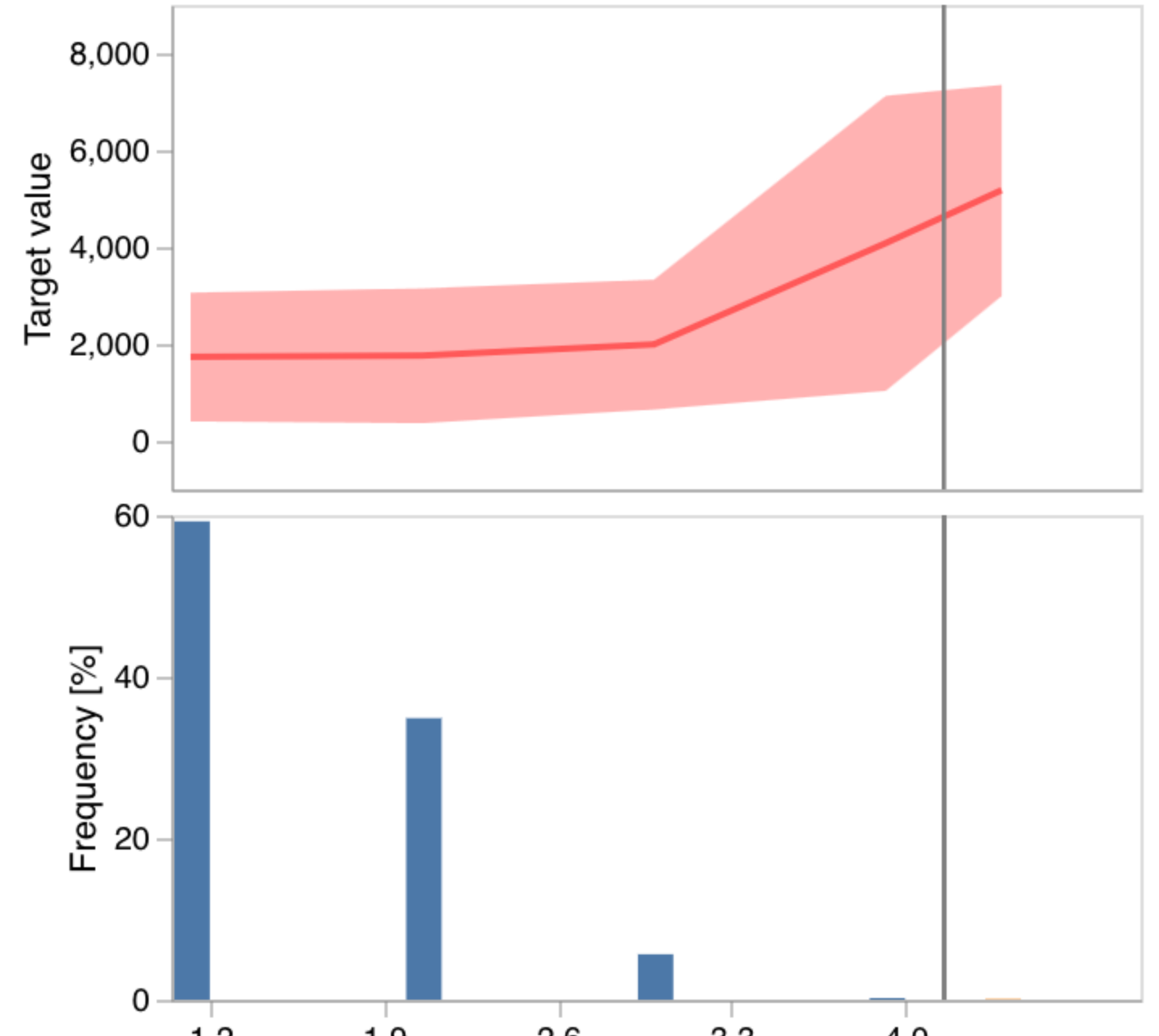
key	value	
Feature name	bathrooms	
Type	numeric	
Prediction power	0.0211	
Valid	100%	
Missing	0%	
Outliers	0.379%	
Min	1	
Max	20	
Mean	1.79	



Histogram of bathrooms with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

deposit · numeric

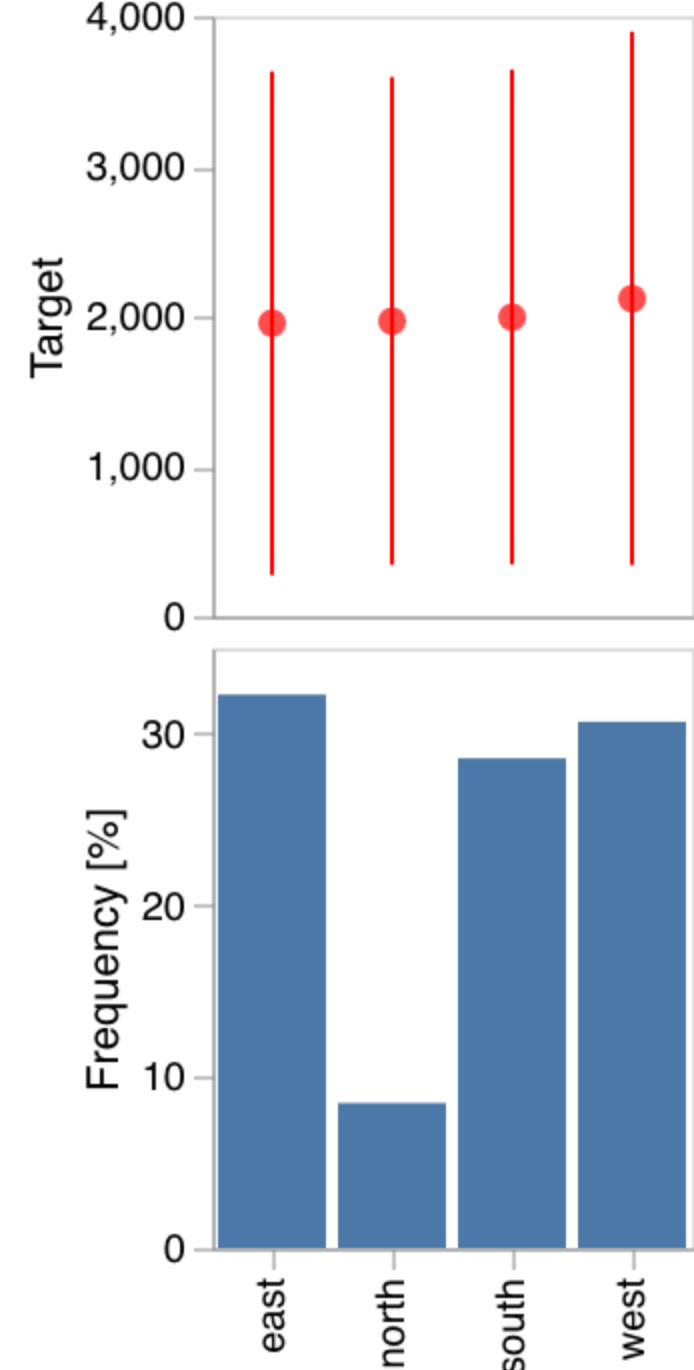
key	value	
Feature name	deposit	
Type	numeric	
Prediction power	0.0201	
Valid	58.6%	
Missing	41.4%	
Outliers	0.074%	
Min	1	
Max	6	
Mean	1.47	



Histogram of deposit with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation. The red bars contain outliers and the value below them is the outliers average.

orientation · categorical

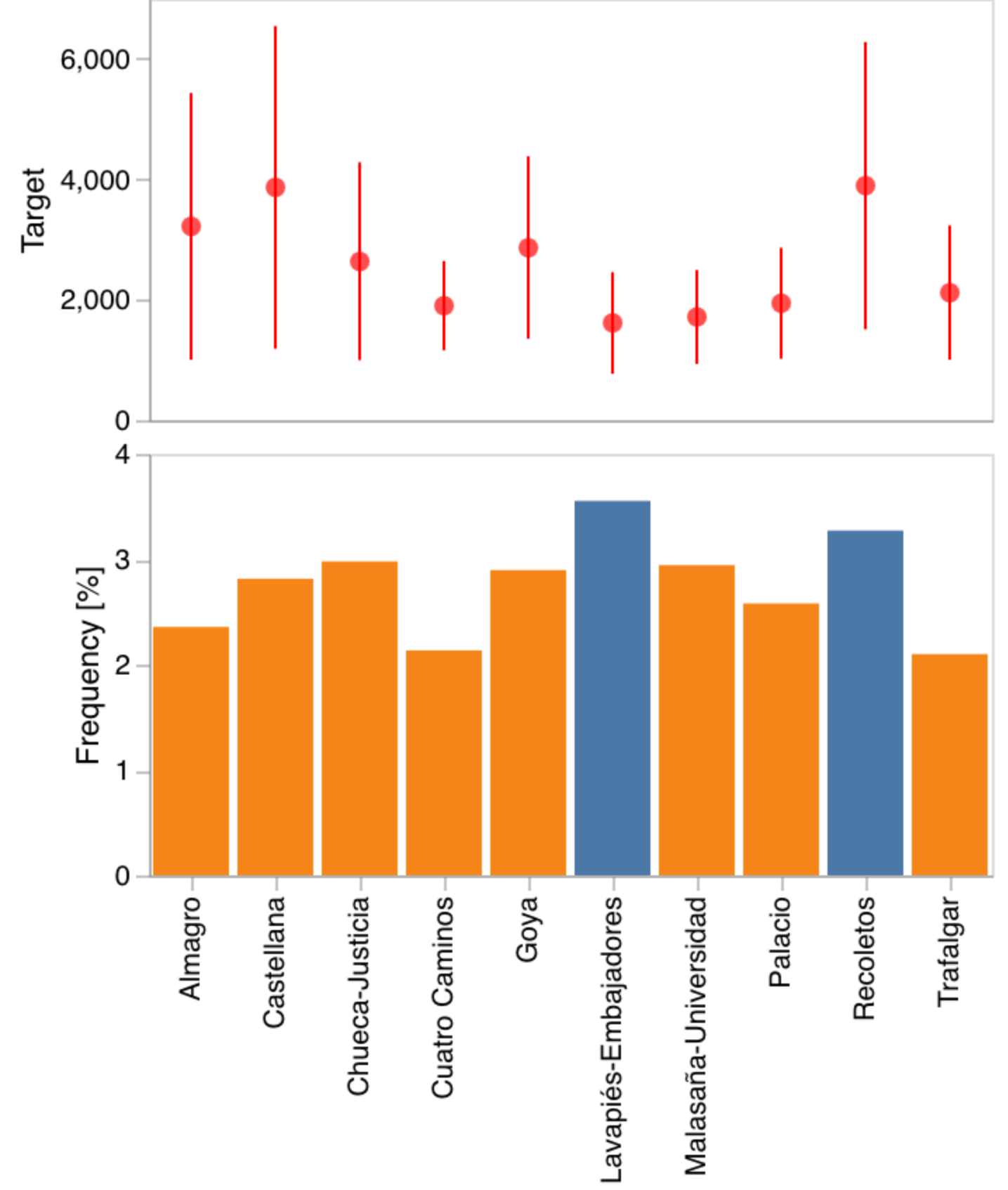
key	value	
Feature name	orientation	
Type	categorical	
Prediction power	0.0128	
Valid	47.8%	
Missing	52.2%	



Histogram of the frequent values of orientation with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

subdistrict · categorical

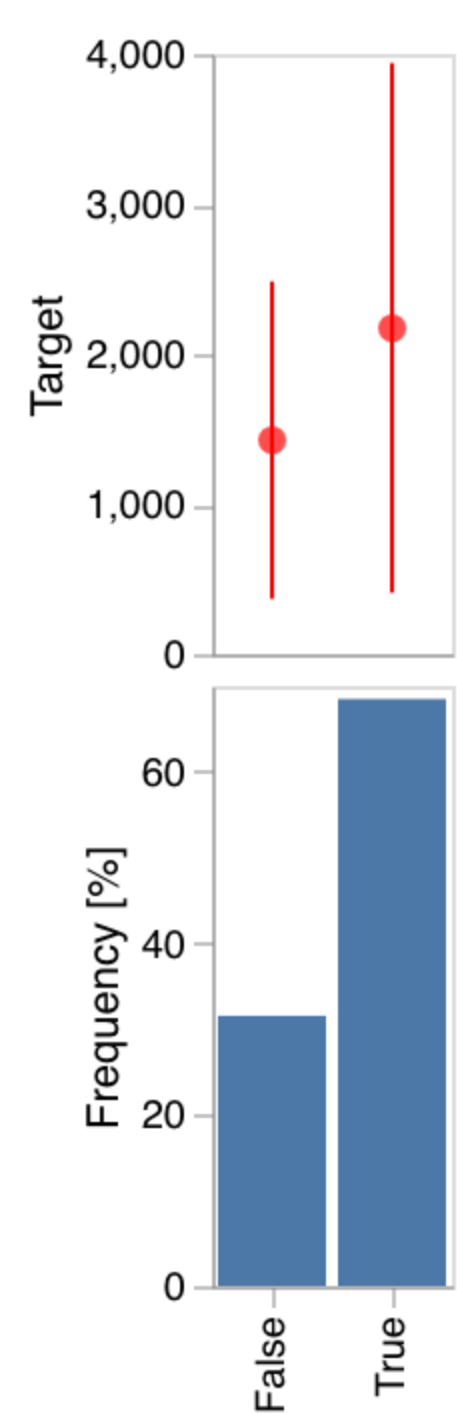
key	value	
Feature name	subdistrict	
Type	categorical	
Prediction power	0.0105	
Valid	92.4%	
Missing	7.6%	



Histogram of the frequent values of subdistrict with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

air_conditioning · binary

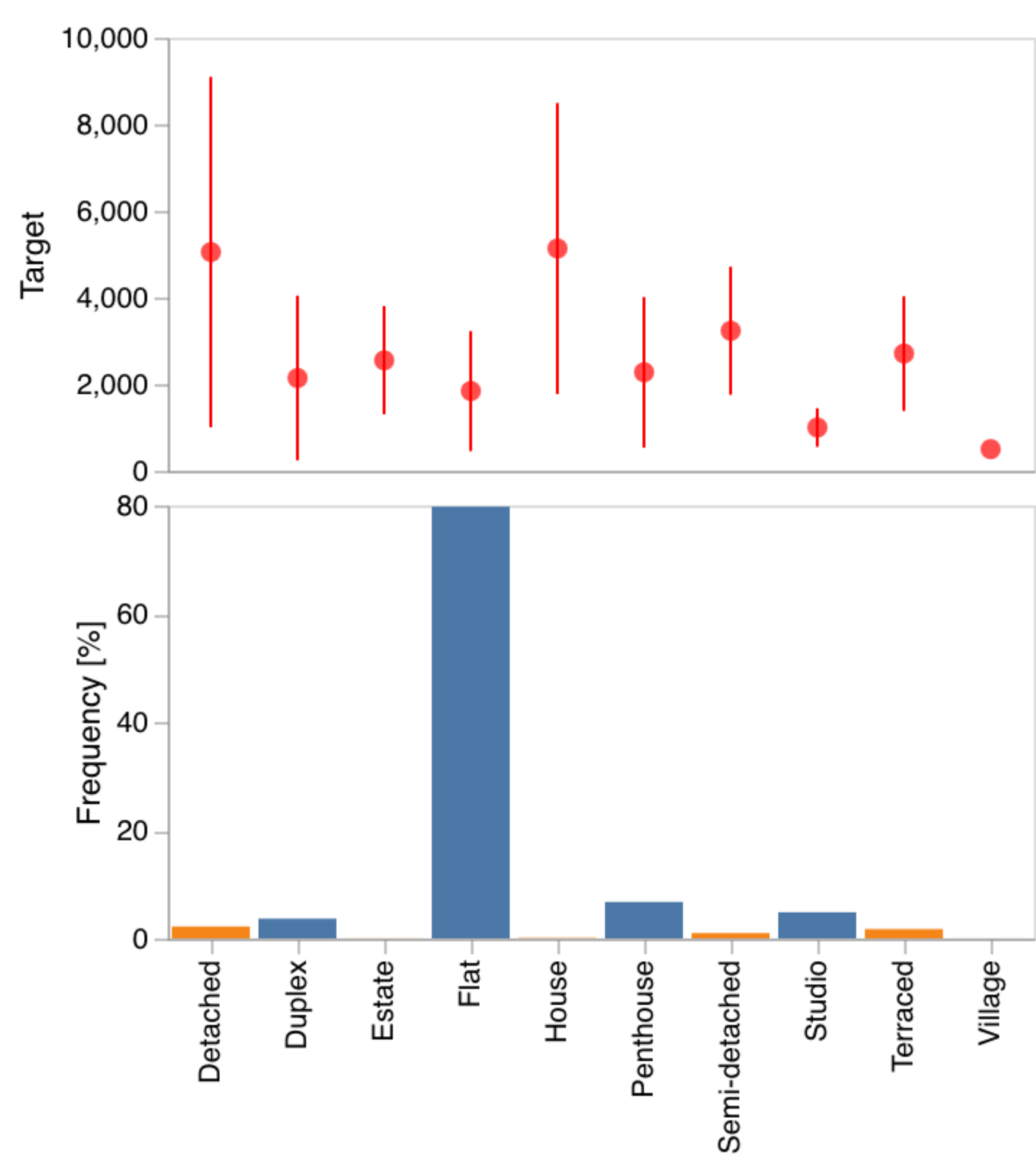
key	value	
Feature name	air_conditioning	
Type	binary	
Prediction power	0.00767	
Valid	100%	
Missing	0%	



Histogram of the frequent values of air_conditioning with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

type · categorical

key	value	
Feature name	type	
Type	categorical	
Prediction power	0.00447	
Valid	100%	
Missing	0%	



Histogram of the frequent values of type with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

Rare classes

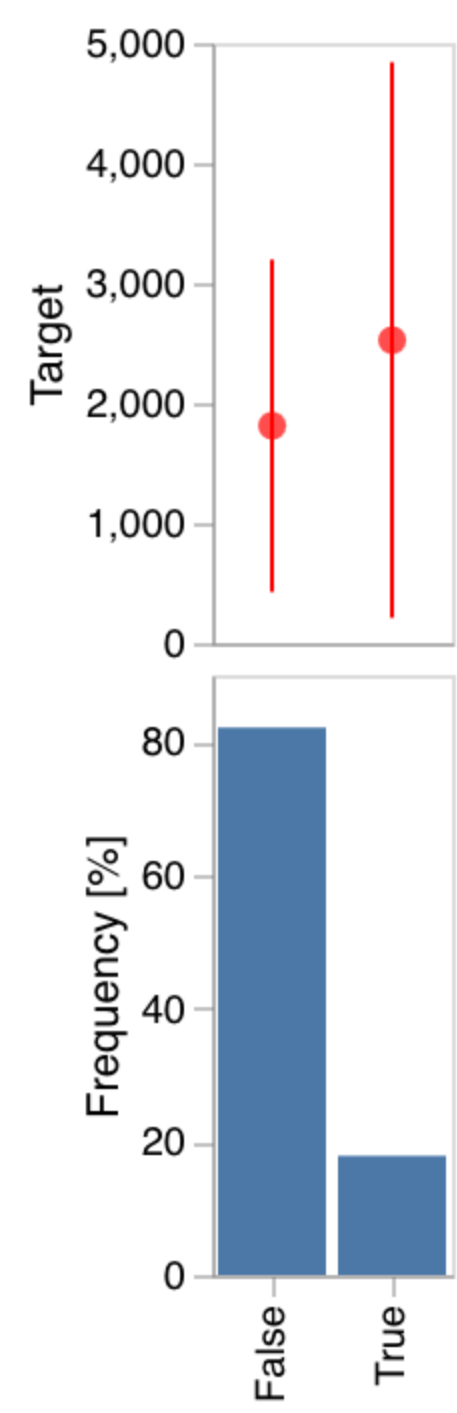
Medium

i

The feature type has 7 categories which are rare: Duplex, Detached, Terraced, Semi-detached, House, Estate, Village. This could lead to overfitting and subpar prediction performance. Consider consolidating them into a single category using the Replace rare values transform under Handle outliers or Find and replace substring transform under Search and edit.

balcony · binary

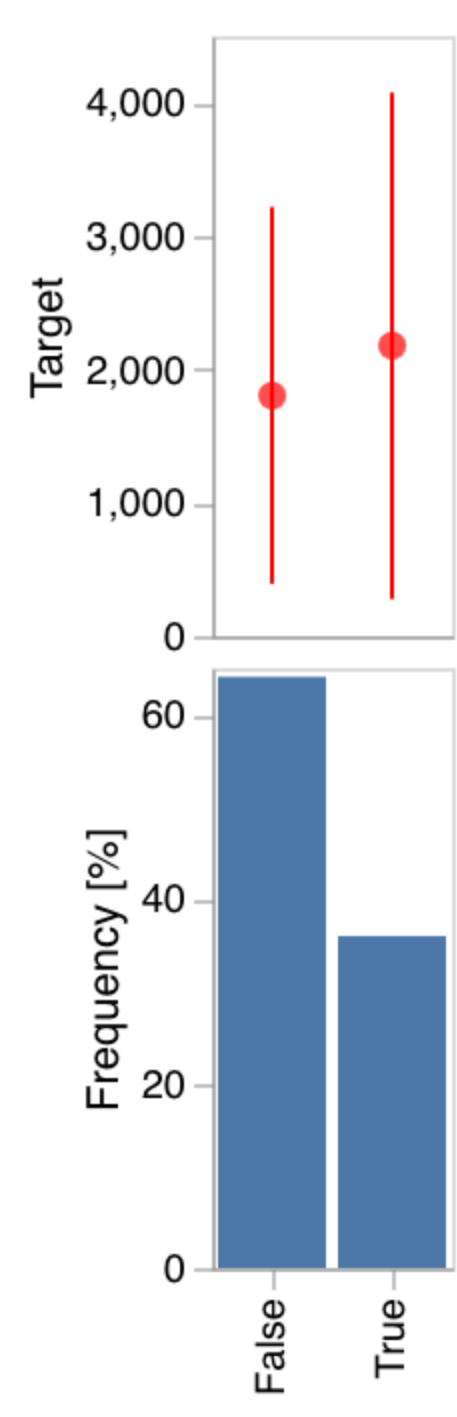
key	value	
Feature name	balcony	
Type	binary	
Prediction power	0.00447	
Valid	100%	
Missing	0%	



Histogram of the frequent values of balcony with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

terrace · binary

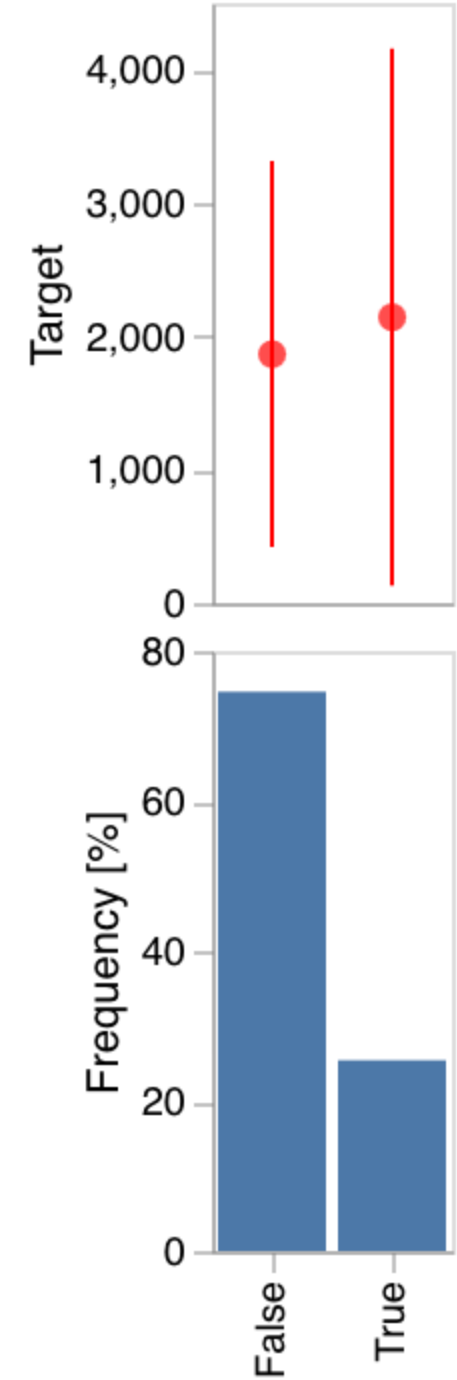
key	value	
Feature name	terrace	
Type	binary	
Prediction power	0.00415	
Valid	100%	
Missing	0%	



Histogram of the frequent values of terrace with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

swimming_pool · binary

key	value	
Feature name	swimming_pool	
Type	binary	
Prediction power	0.00415	
Valid	100%	
Missing	0%	

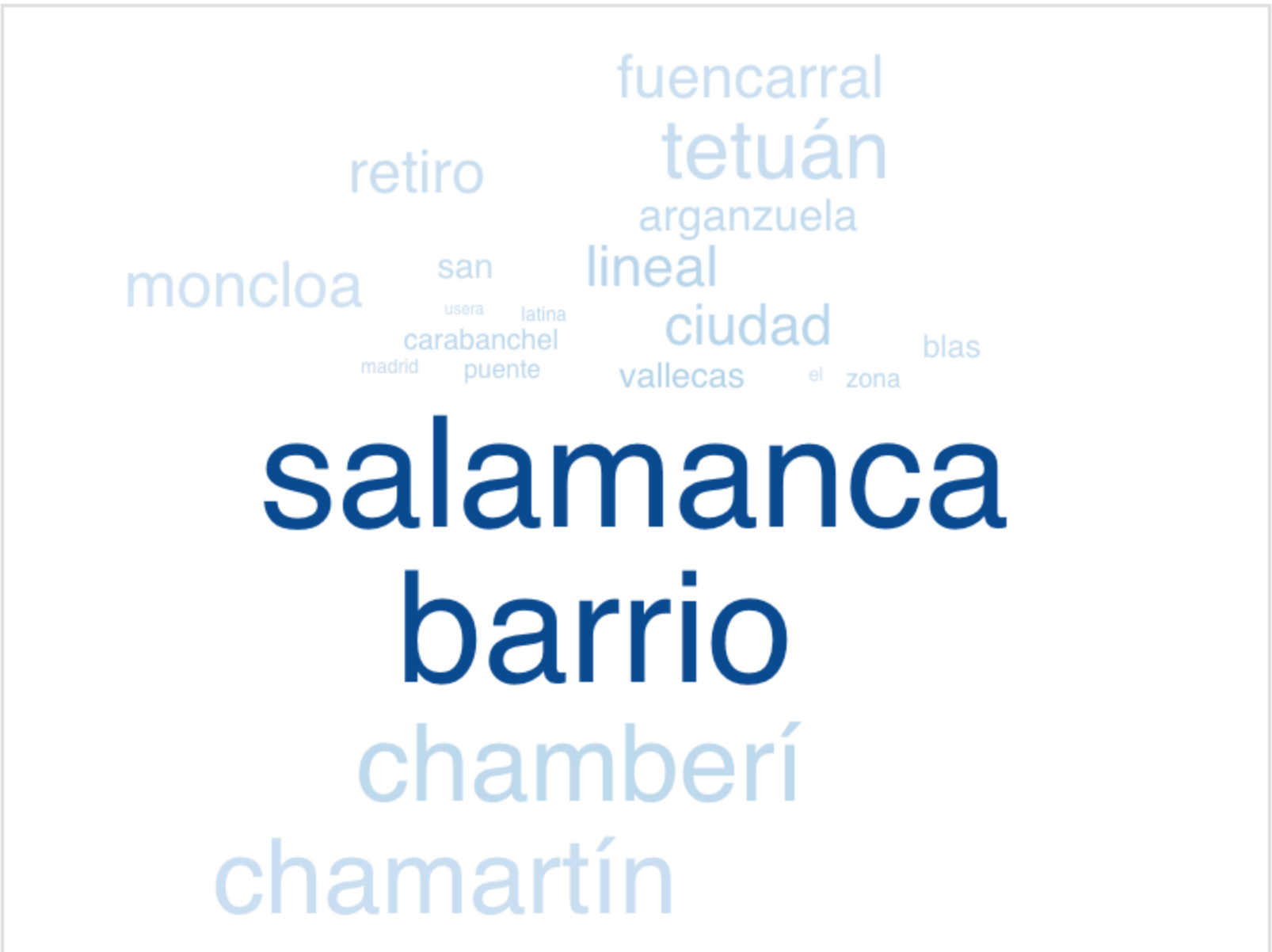


Histogram of the frequent values of swimming_pool with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

district · text

key	value	
Feature name	district	
Type	text	
Prediction power	0.00351	
Valid	100%	
Missing	0%	

Word	Frequency	Predict
salamanca	12.1%	0.071
barrio	12.1%	0.070
finca	0.347%	0.044
prado	0.347%	0.044
somosaguas	0.509%	0.036
vallecas	2.78%	0.010
ciudad	4.53%	0.009

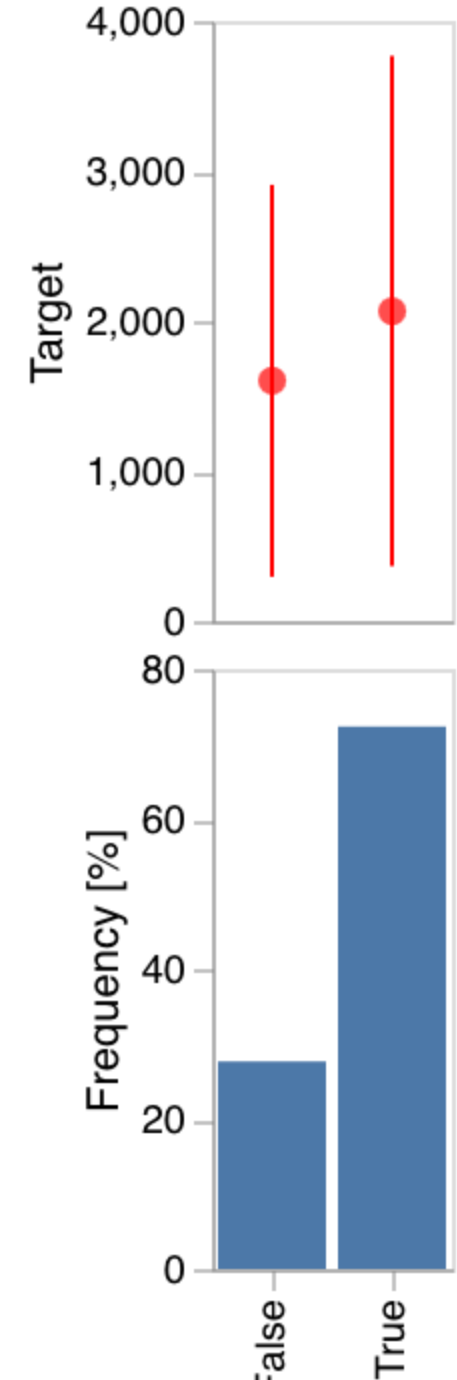


Common words. The size corresponds to word frequency and the color the predictive power. Darker words are more predictive than lighter ones.

No textual patterns found.

fitted_wardrobes · binary

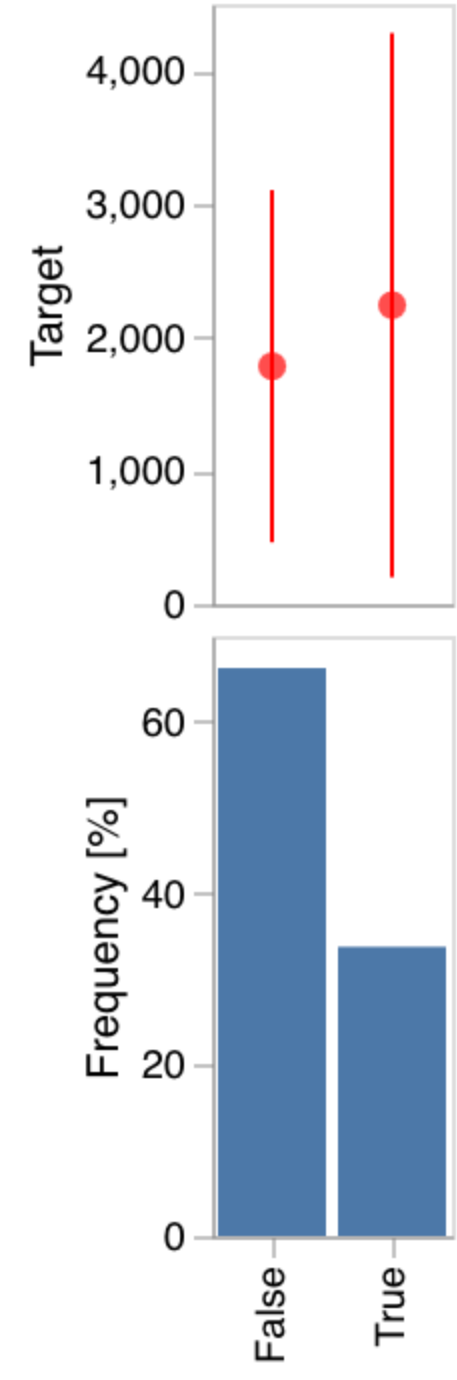
key	value	
Feature name	fitted_wardrobes	
Type	binary	
Prediction power	0.00319	
Valid	100%	
Missing	0%	



Histogram of the frequent values of fitted_wardrobes with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

garage_included · binary

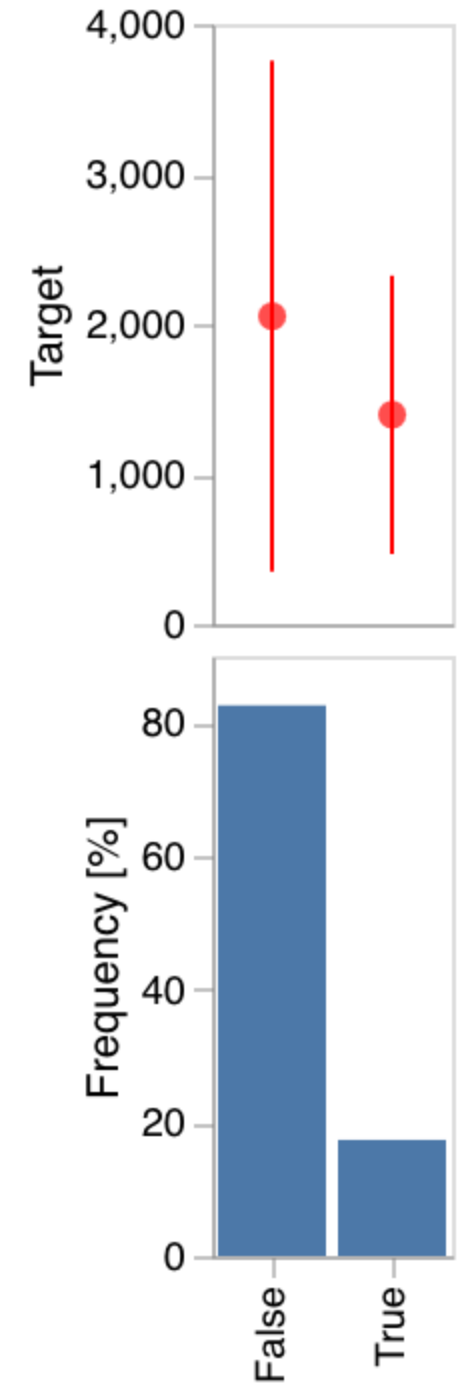
key	value	
Feature name	garage_included	
Type	binary	
Prediction power	0.00288	
Valid	100%	
Missing	0%	



Histogram of the frequent values of garage_included with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

private_owner · binary

key	value	
Feature name	private_owner	
Type	binary	
Prediction power	0.00128	
Valid	100%	
Missing	0%	



Histogram of the frequent values of private_owner with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

url · text

key	value	
Feature name	url	
Type	text	
Prediction power	0.000958	
Valid	100%	
Missing	0%	

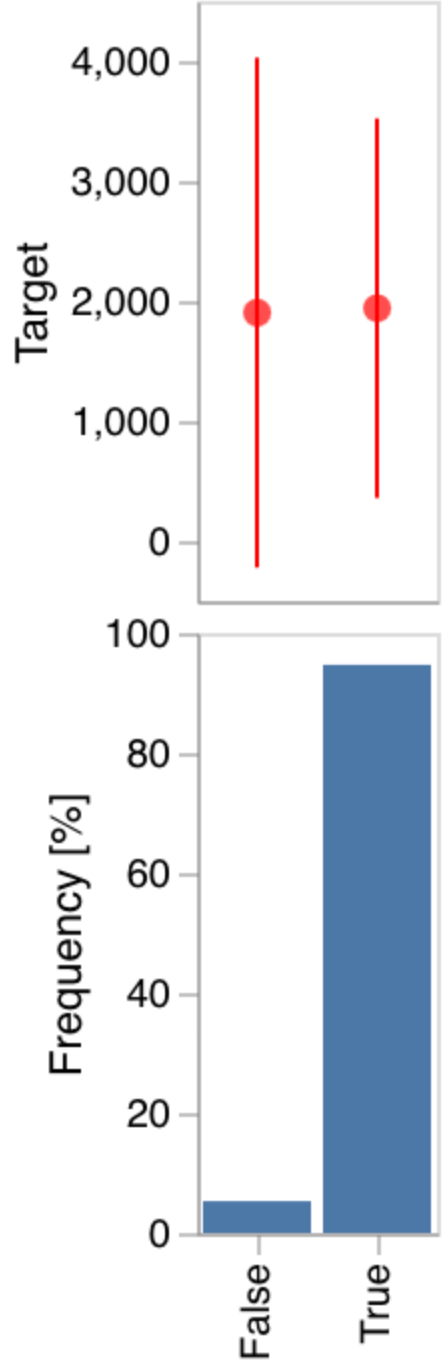
Word	Frequency	Predict
com	100%	0
en	100%	0
https	100%	0
idealista	100%	0
inmueble	100%	0
www	100%	0



Common words. The size corresponds to word frequency and the color the predictive power. Darker words are more predictive than lighter ones.

furnished · binary

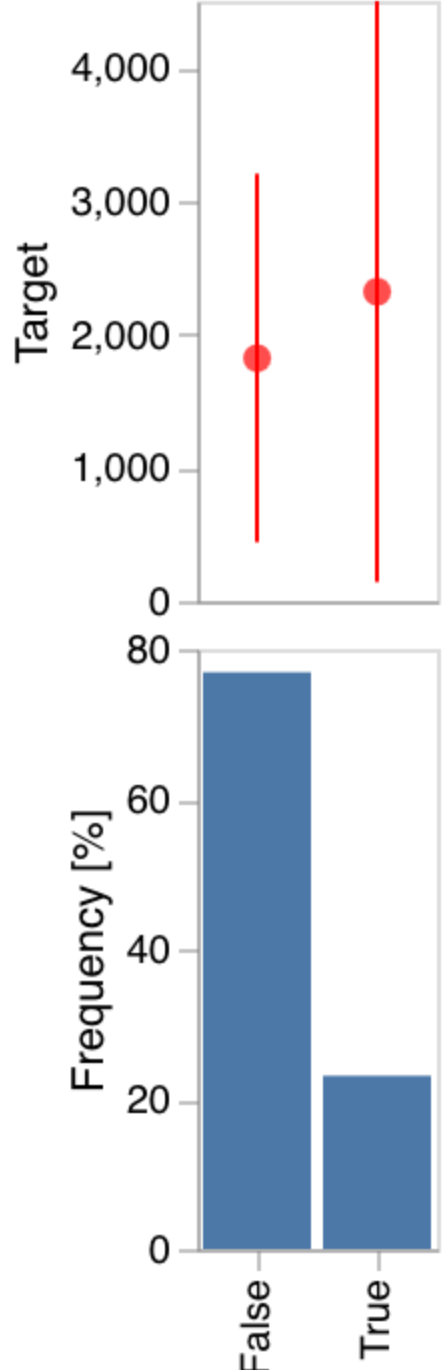
key	value	
Feature name	furnished	
Type	binary	
Prediction power	0.000958	
Valid	100%	
Missing	0%	



Histogram of the frequent values of furnished with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

storeroom · binary

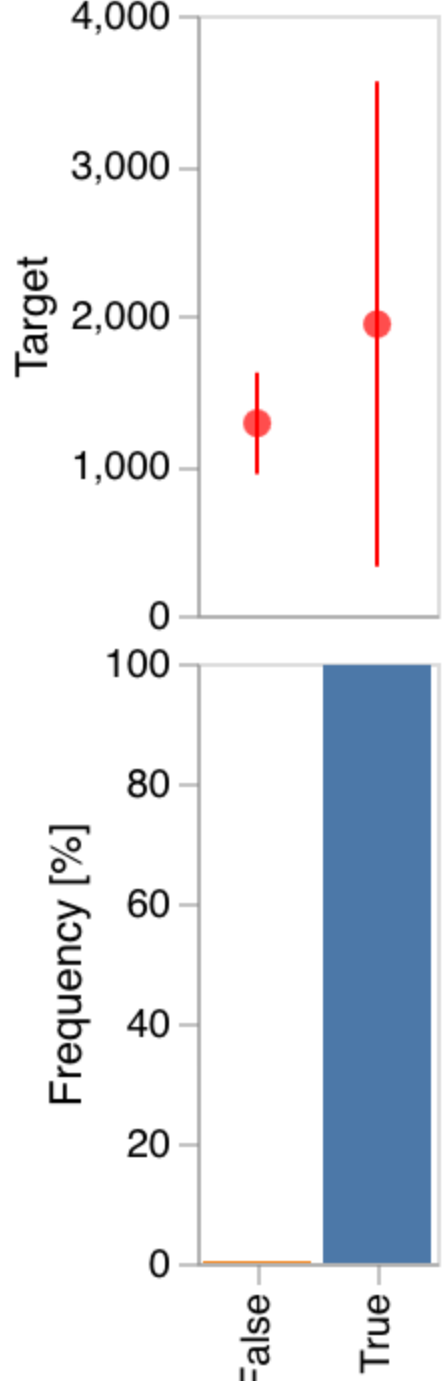
key	value	
Feature name	storeroom	
Type	binary	
Prediction power	0.000958	
Valid	100%	
Missing	0%	



Histogram of the frequent values of storeroom with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

second_hand · binary

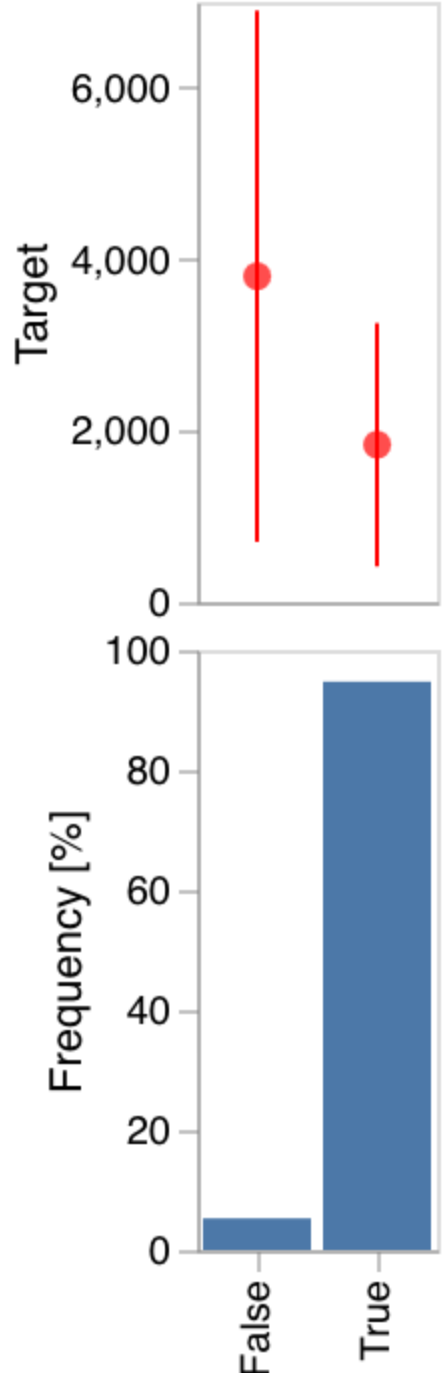
key	value	
Feature name	second_hand	
Type	binary	
Prediction power	0.000639	
Valid	100%	
Missing	0%	



Histogram of the frequent values of second_hand with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

lift · binary

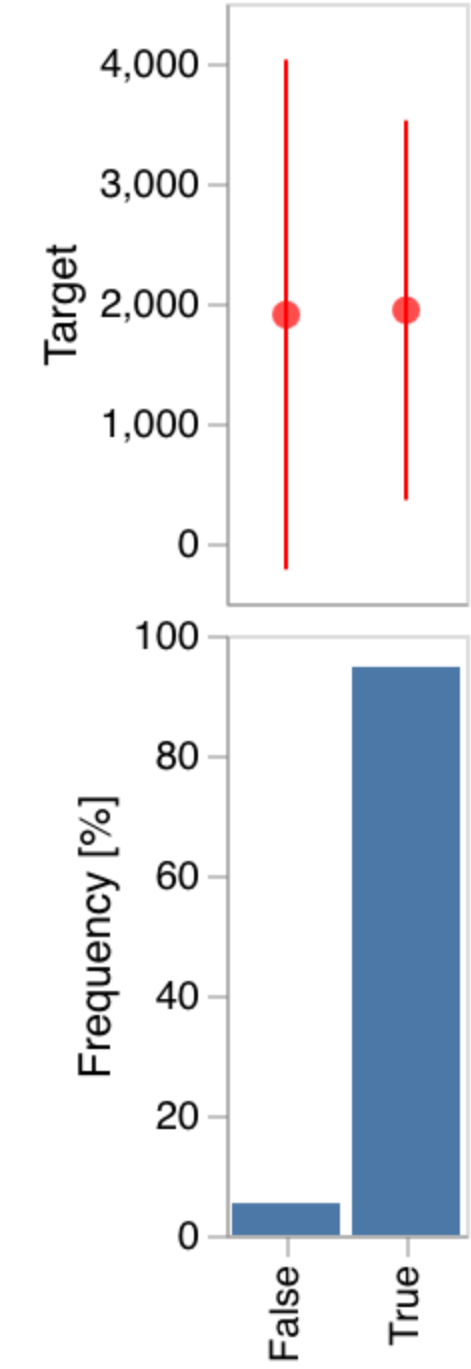
key	value	
Feature name	lift	
Type	binary	
Prediction power	0.000319	
Valid	100%	
Missing	0%	



Histogram of the frequent values of lift with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

equipped_kitchen · binary

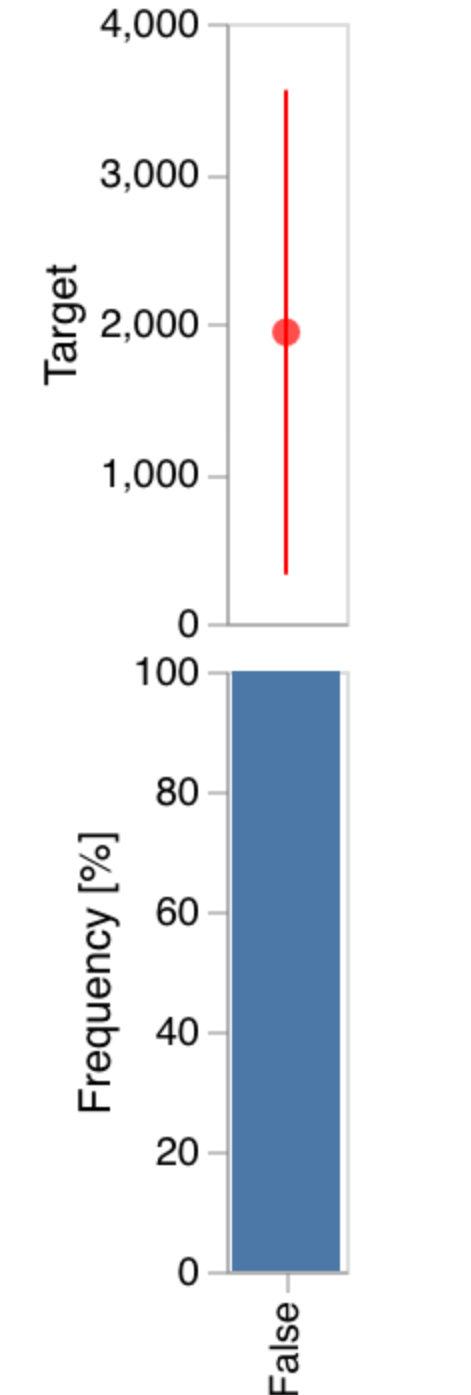
key	value	
Feature name	equipped_kitchen	
Type	binary	
Prediction power	0	
Valid	100%	
Missing	0%	



Histogram of the frequent values of equipped_kitchen with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

garden_area · categorical

key	value	
Feature name	garden_area	
Type	categorical	
Prediction power	0	
Valid	100%	
Missing	0%	



Histogram of the frequent values of garden_area with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target average with a confidence band of one standard deviation.

Definitions

Feature types

Numeric: Numeric values, either floats or integers. For example: age, income. The machine learning models assume that numeric values are ordered and a distance is defined over them, that is 3 is closer to 4 than to 10 and $3 < 4 < 10$.
Categorical: The column entries belong to a set of unique values which is usually much smaller than the number of entries in the column. For example, a column of length 100 containing the unique values "Dog", "Cat" and "Mouse". The values could be numeric, text or combination of both e.g. "Horse", "House", 8, "Love" and 3.1 are all valid values and can be found in the same categorical column. As opposed to numeric features, the machine learning model does not assume order or distance on the values of a categorical feature (even when all the values are numbers).
Binary: A special case of categorical feature where the cardinality of the set of unique values is 2.
Text: A text column contains many non-numeric unique values. In extreme cases, all the elements of the column are unique i.e., no two entries are the same.
Datetime: This column contains date and/or time information.

Feature Statistics

Prediction power

Outliers (in numeric features and regression target): Outliers are detected using two statistics that are robust to outliers: median and robust standard deviation (RSTD). RSTD is derived by clipping the feature values to the range [5 percentile, 95 percentile] and calculating the standard deviation of the clipped vector. All values larger than $\text{median} + 5 * \text{RSTD}$ or smaller than $\text{median} - 5 * \text{RSTD}$ are considered to be outliers.
Skew (in numeric features and regression target): Skew measures the symmetry of the distribution and is defined as the third moment of the distribution divided by the third power of the standard deviation. The skewness of the normal distribution or any other symmetric distribution is zero. Positive values imply that the right tail of the distribution is longer than the left tail and negative values - vice versa. As a thumb rule, a distribution is considered skewed when the absolute value of the skew is larger than 3.
Kurtosis (in numeric features and regression target): Pearson's kurtosis measures the heaviness of the tail of the distribution and is defined as the fourth moment of the distribution divided by the square of the second moment. The kurtosis of the normal distribution is 3. Thus, kurtosis values lower than 3 imply that the distribution is concentrated around the mean and the tails are lighter than the tails of the normal distribution. On the other hand, Kurtosis values higher than 3 imply heavier tails and/or outliers.

Values

Numeric features / regression target: All values that could be casted to finite floats are valid. Missing values are not valid.
Categorical / binary / text features / classification target: All values that are not missing are valid.
Datetime features: All values that could be casted to datetime object are valid. Missing values are not valid.
Invalid values: Either missing or could not be casted to the desired type. See the definition of valid values for more information

Quick model metrics

Regression

R2 (coefficient of determination): R2 is the proportion of the variation in the target that is predicted by the model. R2 is in the range of [-inf, 1] where 1 is the score of the model that predicts the target perfectly and 0 is the score of the trivial model that always predicts the target mean.
MSE: Mean squared error. MSE is in the range [0, inf] where 0 is the score of the model that predicts the target perfectly
MAE: Mean absolute error. MAE is in the range [0, inf] where 0 is the score of the model that predicts the target perfectly
RMSE: Root mean square error. RMSE is in the range [0, inf] where 0 is the score of the model that predicts the target perfectly
Median absolute error: In the range [0, inf] where 0 is the score of the model that predicts the target perfectly

Classification

Accuracy: The ratio of samples that are predicted accurately. Accuracy is in the range [0, 1] where 0 is the score of the model that predicts all samples wrong and 1 is the score of the perfect model.
Balanced accuracy: The ratio of samples that are predicted accurately when class weights are adjusted to balance the data. That is, all classes are given the same importance, regardless of their frequency. Balanced accuracy is in the range [0, 1] where 0 is the score of the model that predicts all samples wrong and 1 is the score of the perfect model.
ROC-AUC: (Binary classification) Area under the receiver operating characteristic curve. ROC-AUC is in the range [0, 1] where a random model will yield a score of 0.5 and the perfect model 1.
ROC-AUC (OVR): (Multi-class classification) Area under the receiver operating characteristic curve calculated separately for each label using one vs rest and the average is reported. ROC-AUC is in the range [0, 1] where a random model will yield a score of 0.5 and the perfect model 1.
Precision: Precision is defined for a specific class. Precision is the fraction of true positives out of all the instances that the model classified as that class. Precision is in the range [0, 1] where 1 is the score of the model that have no false positives for the class. For binary classification, we report the precision of the positive class.
Recall: Recall is defined for a specific class. Recall is the fraction of the relevant class instances that are successfully retrieved. Recall is in the range [0, 1] where 1 is the score of the model that classifies correctly all the instances of the class. For binary classification, we report the recall of the positive class.
F1: F1 is defined for a specific class and it is the harmonic mean of the precision and recall. F1 is in the range [0, 1] where 1 is the score of the perfect model. For binary classification, we report F1 for the positive class.

Pattern Learning

Textual Patterns

Patterns describe the textual format of a string using an easy to read format. The following are examples of textual patterns:

- "{digits:4-7}" describes a sequence of digits of length between 4 and 7.
- "{alnum:5}" describes an alpha-numeric string of length exactly 5.

Canvas infers the patterns by looking at samples of non-empty strings from your data. It can describe many of the commonly used patterns. The relevance (%) indicates how much of the data is estimated to match the pattern. Using the textual pattern, you can see which rows in your data you need to correct or drop.

The following describes the patterns that Canvas can recognize:

Pattern	Textual Format
{alnum}	alphanumeric strings
{any}	any string of word characters
{digits}	a sequence of digits
{lower}	a lowercase word
{mixed}	a mixed-case word
{name}	a word beginning with a capital letter
{upper}	an uppercase word
{whitespace}	whitespace characters

A word character is either an underscore or a character that may appear in a word in any language. For example, the strings 'Hello_world' and 'écouté' both consist of word characters. 'H' and 'é' are both examples of word characters.